
Analog Electronics Lab

Laboratory Manual

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Corequisites	PBEET304

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Preface

This manual has been prepared to accompany the laboratory course **Analog Electronics Lab (PCEEL308)(2024 scheme)**, offered in Semester 3 of the B.Tech. programme in Electrical and Electronics Engineering. The course is designed to provide hands-on exposure to the design, simulation, and hardware realisation of fundamental analog electronic circuits employing diodes, BJTs, JFETs, MOSFETs, operational amplifiers, and the 555 timer IC.

Each experiment in this manual follows a uniform structure consisting of the aim, theoretical background, pre-lab assignment, components and equipment required, circuit diagram, design procedure, experimental procedure, expected/model graphs, observation tables, results, and viva-voce questions. Students are expected to complete the pre-lab assignment prior to each laboratory session and to record all readings directly in the laboratory record during the conduct of the experiment.

Course Information

Course Objectives

1. To design transistor and operational-amplifier based electronic circuits.
2. To simulate and implement the designed circuits in hardware and validate experimental results against theoretical/simulated predictions.

Course Outcomes

On successful completion of this course, the student will be able to:

CO	Course Outcome	Bloom's Level
CO1	Use the various electronic instruments for conducting experiments.	K3
CO2	Design and develop various electronic circuits using diodes and Zener diodes.	K3
CO3	Design and implement amplifier and oscillator circuits using BJT and JFET.	K3
CO4	Design and implement basic circuits using IC (OPAMP and 555 timer).	K3
CO5	Simulate electronic circuits using circuit simulation software.	K3
CO6	Use PCB layout software for circuit design.	K3

Table 1: Course Outcomes and Bloom's Knowledge Level

K1: Remember, K2: Understand, K3: Apply, K4: Analyse, K5: Evaluate, K6: Create

Scheme of Evaluation

Continuous Internal Evaluation (CIE – 50 Marks)

Component	Marks
Attendance	5
Preparation / Pre-Lab Work, Viva, and Timely Completion of Lab Reports / Record (Continuous Assessment)	25
Internal Examination	20
Total	50

Table 2: CIE Marks Distribution

The continuous assessment component (25 marks) is further broken down as:

- **Preparation and Pre-Lab Work (7 Marks):** Pre-lab assignments/quizzes and understanding of theoretical background.
- **Conduct of Experiments (7 Marks):** Adherence to procedure, accurate execution, safety, skill proficiency, and teamwork.
- **Lab Reports and Record Keeping (6 Marks):** Clarity, completeness, accuracy, and timely submission.
- **Viva Voce (5 Marks):** Ability to explain the experiment, results, and underlying principles.

The final marks for preparation, conduct, viva, and record are computed as the average across all experiments listed in the syllabus.

End Semester Examination (ESE – 50 Marks)

Component	Marks
Procedure / Preliminary Work / Design / Algorithm	10
Conduct of Experiment / Execution of Work / Troubleshooting / Programming	15
Result with Valid Inference / Quality of Output	10
Viva Voce	10
Record	5
Total	50

Table 3: ESE Marks Distribution

- Students shall be permitted to appear for the end-semester examination only upon submission of a duly certified record.
- The external examiner shall endorse the record at the time of the examination.

CO–PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	2	3	3	3	3				3	3		
CO3	2	3	3	3	3				3	3		
CO4	2	3	3	3	3				3	3		
CO5	2	3	3	3	3				3	3		
CO6	3								3	3		

Table 4: CO–PO Mapping (1: Slight, 2: Moderate, 3: Substantial, blank: No Correlation)

Text Books and References

1. Robert T. Paynter, *Introductory Electronic Devices and Circuits*, Pearson Education.
2. Boylestad R. L. and Nashelsky L., *Electronic Devices and Circuit Theory*, Pearson Education.
3. Donald A. Neamen, *Electronic Circuits: Analysis and Design*, McGraw Hill.

General Instructions to Students

Laboratory Conduct

1. Students must bring the laboratory record, observation notebook, graph sheets, and a scientific calculator to every session.
2. The pre-lab assignment for the experiment scheduled for the day must be completed and ready for evaluation before entering the laboratory.
3. Students shall enter the laboratory only with permission and shall occupy the workbench allotted to them.
4. Circuit connections must be checked and approved by the faculty-in-charge or laboratory instructor before the power supply is switched on.
5. Readings must be entered directly into the observation notebook at the time of the experiment; fabricated or estimated readings will not be accepted.
6. Calculations, graphs, and inferences must be completed and the experiment got corrected on the same day, wherever possible, or within the stipulated time announced by the instructor.
7. Students must simulate the circuit using the prescribed circuit simulation software prior to hardware implementation, wherever applicable.
8. Mobile phones shall be kept in silent mode and shall not be used during the conduct of the experiment, except as directed for simulation work.
9. Any damage to equipment or components due to negligence shall be reported immediately and may be subject to the institution's equipment-damage policy.
10. Students are responsible for returning all components, probes, and patch cords to their designated locations and for switching off all instruments and the workbench power supply before leaving the laboratory.

Best Practices for Breadboarding

Breadboarding is a fundamental skill in analog electronics. A neat and disciplined approach in breadboarding will help achieving the expected results without any hassle. Following these best practices will help you build reliable circuits and simplify troubleshooting:

- **Plan Your Layout:** Before placing components, mentally (or physically with a drawing) plan the layout of your circuit on the breadboard. Group related components.
- **Shortest Connections:** Use short, neat wires for connections. Long, messy wires can introduce noise, capacitance, and inductance, and make troubleshooting difficult.

- **Color Coding:** Use different colored wires for different purposes (e.g., red for +Vcc, black for ground, blue for signal lines). This greatly aids in tracing connections.
- **Power Rails:** Always connect the power supply rails (+Vcc and GND) to the dedicated power buses on the breadboard.
- **Component Orientation:** Pay close attention to the polarity of components like diodes, LEDs, and electrolytic capacitors.
- **Secure Connections:** Ensure all component leads and wires are firmly inserted into the breadboard holes. Loose connections are a common cause of circuit malfunction.
- **Start Simple:** When troubleshooting, simplify your circuit. Isolate sections and test them individually.
- **Component Identification:** Familiarize yourself with common component packages and their pinouts (e.g., op-amp pin configurations, transistor pinouts).
- **Avoid Overcrowding:** Don't try to cram too many components into a small area. Use a larger breadboard if necessary.
- **Check Continuity:** Before applying power, use a Digital Multimeter (DMM) to check for short circuits between power rails and to verify critical connections.

Observations and Data Recording

Accurate and detailed observations are crucial for a successful lab experience and for writing comprehensive lab reports.

- **Lab Notebook:** Maintain a dedicated lab notebook.
- **Date and Time:** Always start each entry with the date and time.
- **Experiment Title:** Clearly state the experiment title and number.
- **Circuit Diagrams:** Draw clear and labeled circuit diagrams.
- **Component Values:** Record all component values used in your circuit.
- **Measurements:** Tabulate all measured data with appropriate units.
- **Waveforms:** Sketch or capture screenshots of oscilloscope waveforms, labeling axes, amplitudes, and frequencies.
- **Unexpected Results:** Note down any unexpected observations or deviations from expected results. This is often where valuable learning occurs.
- **Troubleshooting Steps:** Document any troubleshooting steps taken and their outcomes.
- **Inferences and Discussions:** Write down your inferences and discuss the implications of your observations.

Lab Report (Record) Submission

1. The first thing to note is that the lab report/record is **NOT A COPY** of the lab manual. The laboratory record is, as its name indicate, a record/report of the experiment done in the lab. The lab manual may include information required for aiding the writing of the report.
2. The lab record must contain, for each experiment: aim, components/equipment list, design, circuit diagram (with all component labels/values clearly shown), experimental procedure, observation tables, model graphs, sample calculations, result, and inferences. Please note that it is not required to write the theory and informational notes that appear in this lab manual.
3. It is suggested to draw the circuit diagrams and observation tables on the left side of the record, and preferably **not on a graph page**. Graph pages are for drawing graphs, waveforms and sample/model curves/graphs.
4. Records must be certified by the faculty-in-charge within one week of completion of the experiment.
5. Submission of a duly certified record is mandatory for eligibility to appear for the end-semester examination.

Evaluation of Pre-Lab Assignments

Two pre-lab assignments are prescribed by the syllabus prior to the regular experiments:

1. Measurement of current, voltage, frequency, and phase shift of a signal in an RC network using a cathode-ray oscilloscope (CRO) / digital storage oscilloscope (DSO).
2. Introduction to circuit simulation using a circuit simulation software package (e.g., LTspice, PSpice, Multisim, or NgSpice/QUCS).

Students are expected to complete these foundational exercises before attempting the numbered experiments listed in Table 6.

Safety Guidelines

Safety is paramount in any laboratory environment. Please read and adhere to the following safety guidelines:

Safety Note

1. **No Food or Drink:** Do not bring or consume food or drinks in the lab.
2. **Work Area:** Keep your workspace clean and organised. Cluttered benches can lead to accidents.
3. **Before Power On:** Always verify the power supply voltage and polarity before connecting it to the circuit. Reversed polarity can permanently damage diodes, transistors, op-amps, and ICs.
4. **Power Off First:** Switch off the power supply before making, modifying, or removing any circuit connections.
5. **Component Handling:** Handle components carefully. Be aware of the polarity of diodes, LEDs, and electrolytic capacitors. Do not exceed the maximum voltage, current, or power ratings of any active or passive component. Always compute and verify current-limiting resistor values before energising a circuit. Use current-limiting resistors in series with LEDs, diodes, and Zener diodes during testing to prevent excessive current flow. Electrolytic capacitors are polarised. Verify polarity before connecting; reversed polarity may cause the capacitor to overheat, vent, or rupture.
6. **Tool Usage:** Use tools (wire strippers, pliers, screwdrivers) correctly and for their intended purpose.
7. **No Short-circuits:** Avoid short-circuiting the output terminals of a regulated DC power supply or the output of an operational amplifier.
8. **Hot Components:** Be aware of the heat-sink and case temperature of power transistors and voltage regulators during prolonged operation; avoid direct contact.
9. Do not work on live circuits with wet hands or in a wet environment.
10. **Report Accidents:** In the event of a component overheating, emitting smoke, or unusual odour, immediately switch off the power supply and inform the laboratory instructor before attempting further investigation. Immediately report any accidents, spills, or equipment malfunctions to the lab instructor or faculty-in-charge or TA.
11. **No Horseplay:** Maintain a professional demeanour and avoid any behaviour that could endanger yourself or others.

12. **Emergency:** Be aware of the location of emergency exits and first-aid kits. Familiarise yourself with the location of the mains isolation switch and the fire extinguisher in the laboratory.
13. **Before Leaving:** Ensure your workstation is clean and all equipment is turned off before leaving the lab.

Precautions

1. Identify the pin configuration (pinout) of every IC and transistor from its datasheet before inserting it into the breadboard.
2. Avoid loose or floating connections, particularly at the input terminals of operational amplifiers, since floating CMOS/op-amp inputs may result in unpredictable circuit behaviour.
3. Keep signal leads and connecting wires as short as practicable to minimise stray capacitance and inductance, particularly in high-frequency oscillator circuits.
4. Re-verify all connections against the circuit diagram before switching on the power supply, and recheck after every modification.
5. Handle CRO/DSO probes with care; ensure the ground (common) lead of the probe is connected to the circuit ground/common point to avoid ground loops and inaccurate measurements.

List of Equipment and Software

Introduction to Essential Lab Equipment

Familiarity with the following equipment is essential for performing experiments in this lab.

Digital Storage Oscilloscope (DSO)

- **Purpose:** The DSO is used to visualize and analyze time-varying electrical signals (waveforms). It displays voltage as a function of time.
- **Key Controls:**
 - **Vertical (Volts/Div):** Controls the voltage scale on the screen (voltage per vertical division).
 - **Horizontal (Sec/Div):** Controls the time scale on the screen (time per horizontal division).
 - **Trigger:** Stabilizes repetitive waveforms on the screen. Adjust trigger level and mode (edge, auto, normal).
 - **Channels:** DSOs typically have multiple input channels (CH1, CH2, etc.) to view multiple signals simultaneously.
 - **Cursors/Measurements:** Use cursors to precisely measure voltage and time intervals. Most DSOs have built-in automated measurement functions (e.g., V_{pp} , V_{rms} , frequency, period).
- **Usage Tips:**
 - Always use a proper probe (1X or 10X). Understand the attenuation factor. Calibrate probes if necessary.
 - Ground connections are crucial for accurate measurements.
 - Start with auto-set if you're unsure of the signal characteristics, then fine-tune.

Function Generator

- **Purpose:** Generates various standard waveforms (sine, square, triangle, pulse) at specified frequencies and amplitudes.
- **Key Controls:**
 - **Waveform Selection:** Selects the desired output waveform.
 - **Frequency:** Sets the output frequency.
 - **Amplitude:** Sets the peak-to-peak voltage of the output waveform.
 - **DC Offset:** Adds a DC voltage component to the output signal.
 - **Output Impedance:** Typically 50 Ω . Be aware of impedance matching.

Usage Tips:

- Connect the output of the function generator to your circuit's input.
- Always check the output on the oscilloscope before connecting to your circuit to ensure it's set correctly.

DC Power Supply

- **Purpose:** Provides stable DC voltage and current to power your circuits.
- **Key Controls:**
 - **Voltage Adjustment:** Sets the output voltage.
 - **Current Limit:**¹ Sets/limits the maximum current the supply can deliver. This is a crucial safety feature to protect your circuit from excessive current.
 - **Output On/Off:** Turns the power supply output on or off.

Usage Tips:

- Always set the voltage and current limit before connecting to your circuit. Start with a lower voltage and gradually increase if needed.
- If the current limit indicator lights up, it means your circuit is drawing more current than allowed, possibly indicating a short circuit or fault. Turn off power immediately and troubleshoot.

Digital Multimeter (DMM)

- **Purpose:** Measures various electrical parameters like voltage (AC/DC), current (AC/DC), resistance, and continuity.
- **Key Functions:**
 - **Voltage Measurement (V):** Connect in parallel with the component or circuit portion you want to measure. Select AC/DC option appropriately.
 - **Current Measurement (A):** Connect in series with the circuit. This requires breaking the circuit. Be cautious about the current range.
 - **Resistance Measurement (Ω):** Measure resistance of disconnected components. Never measure resistance in an active circuit.
 - **Continuity Test:** Beeps if there is a low-resistance path (useful for checking connections).

Usage Tips:

- Always select the correct function (V, A, Ω) and range before making a measurement.
- For current measurements, always connect the DMM in series. Incorrect connection can blow the internal fuse.
- Ensure the circuit is de-energized when measuring resistance.

¹Note that by adjusting this control, one cannot vary the current. The current drawn from a voltage-source power supply is decided by the load resistance.

Equipment / Software / Components	Typical Use in this Lab
Regulated DC Power Supply (dual/triple output)	Biasing of transistor, op-amp, and 555 timer circuits
Function / Signal Generator	Input excitation for amplifiers, filters, and frequency-response studies
Digital Storage Oscilloscope (DSO)	Measurement of waveforms, amplitude, frequency, and phase shift
Digital Multimeter (DMM)	DC bias point and resistance/voltage/current measurements
Breadboard and connecting wires	Circuit prototyping
Active and Passive Components (diodes, Zener diodes, BJTs, JFETs, MOSFETs, resistors, capacitors, inductors)	Circuit construction across experiments
Operational Amplifier ICs (e.g., $\mu A741$, LM358)	Op-amp based experiments
Comparator IC (e.g., LM311)	Comparator and Schmitt trigger experiments
555 Timer IC	Astable and monostable multivibrator experiments
Circuit Simulation Software (e.g., LTspice / Qspice / NgSpice-QUCS)	Pre-implementation simulation of all designed circuits
PCB Layout Software (e.g., KiCad)	Introduction to PCB design

Table 5: Indicative list of equipment and software used in the course

List of Experiments

Table 6: List of Experiments – Analog Electronics Lab (PCEEL308)

Expt. No.	Title	CO
PL-1	Measurement of current, voltage, frequency, and phase shift of a signal in an RC network using a CRO/DSO (Pre-Lab)	CO1
PL-2	Introduction to circuit simulation using a circuit simulation software (Pre-Lab)	CO5
1	Clipping and clamping circuits using diodes	CO2
2	Basic RC circuits – High-pass and Low-pass filters	CO1
3	RC-coupled amplifier using BJT in CE configuration – Measurement of gain, bandwidth, and plotting of frequency response	CO3
4	Emitter follower amplifier	CO3
5	JFET amplifier – Measurement of gain, bandwidth, and plotting of frequency response	CO3
6	MOSFET amplifier	CO3
7	Design and testing of voltage regulators – Zener and series	CO2
8	Design and setup of inverting and non-inverting amplifiers	CO4
9	Op-amp circuits – Scale changer, adder, integrator, and differentiator	CO4
10	Precision rectifier using op-amp	CO4
11	Op-amp oscillators – RC phase-shift and Wien bridge oscillator	CO4
12	Op-amp oscillator – LC oscillator (Colpitts or Hartley)	CO4
13	Waveform generation – Square, triangular, and sawtooth waveform generation using op-amps	CO4
14	Basic comparator and Schmitt trigger circuits using op-amp (using comparator ICs such as LM311)	CO4
15	Active filters (one each of high-pass and low-pass)	CO4
16	Instrumentation amplifier	CO4
17	Astable and monostable multivibrator circuits using 555 timer IC	CO4
18	Introduction to PCB layout software	CO6

Format Followed for Each Experiment

Each experiment in the chapters that follow is presented under the following uniform headings:

1. **Aim**
2. **Pre-Lab Assignment**

3. **Components and Equipment Required**
4. **Theory**
5. **Circuit Diagram**
6. **Design**
7. **Procedure**
8. **Precautions**
9. **Model Graph / Expected Waveform**
10. **Observation Table**
11. **Result**
12. **Questions**

The students are expected to submit the report of each experiment, containing the following:

1. **Aim**
2. **Pre-Lab Assignment**
3. **Components and Equipment Required**
4. **Circuit Diagram**
5. **Design**
6. **Procedure**
7. **Model Graph / Expected Waveform**
8. **Observation Tables**
9. **Results/Conclusions**

Pre-Lab 1:

Measurement of Voltage, Frequency and Phase Shift in an RC Network using a Digital Storage Oscilloscope

Objectives

- (i) To become familiar with the front-panel controls of a basic (20–50 MHz) Digital Storage Oscilloscope (DSO) and a function generator.
- (ii) To measure the amplitude (peak, peak-to-peak, and RMS voltage) and frequency of a sinusoidal signal using the DSO.
- (iii) To study the response of a simple series RC network to a sinusoidal input.
- (iv) To measure the phase shift introduced by the RC network between the input and output voltages, using both the **time-difference method** and the **Lissajous (X–Y) method**.
- (v) To compare the experimentally measured phase shift and gain with the theoretically predicted values.

Pre-Lab Theory

The Series RC Network

Consider the series RC circuit shown in Figure 1, driven by a sinusoidal source $v_{in}(t) = V_m \sin(\omega t)$, where the output is taken across the capacitor. This simple linear, passive, circuit has several practical applications. This circuit is often called an *integrator* or a *low-pass filter*. The circuit when given a sinusoidal signal input, outputs a sinusoidal signal, but with different amplitude and phase, depending on the values of R and C .

The transfer function² of this network is

$$H(j\omega) = \frac{V_{out}}{V_{in}} = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC}. \quad (1)$$

It may be noted that this can be obtained by sinusoidal analysis of the network considering R and C as impedances, and using voltage-divider formula.³ Defining the *cutoff (corner) frequency*

$$f_c = \frac{1}{2\pi RC}, \quad (2)$$

²The idea of transfer function is introduced in the course Circuits and Networks. In simple terms, it is the ratio of output to input

³ $V_{Z_2} = V_{in} \frac{Z_2}{Z_1 + Z_2}$

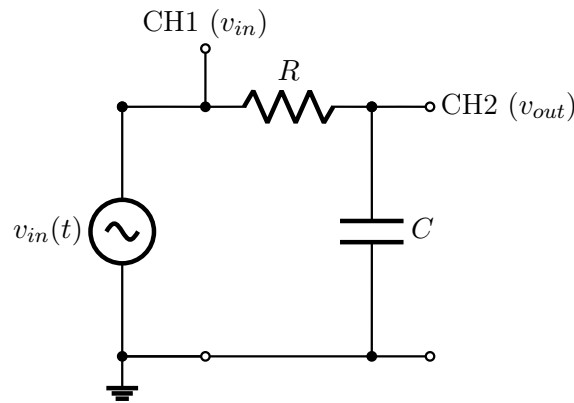


Figure 1: Series RC low-pass network. CH1 of the DSO monitors the source voltage $v_{in}(t)$ and CH2 monitors the voltage across the capacitor $v_{out}(t)$. Both channels share a common ground, which is same as the reference ground of the source.

the magnitude and phase of $H(j\omega)$ are

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (f/f_c)^2}}, \quad (3)$$

$$\phi(f) = -\tan^{-1}\left(\frac{f}{f_c}\right). \quad (4)$$

The cut-off frequency f_c is an important parameter that decides the operating characteristics of this circuit. The circuit behaves differently to signals with frequencies above and below f_c .

Equation (3) shows that the network attenuates the input as frequency increases (a low-pass filter), while Equation (4) shows that $v_{out}(t)$ lags $v_{in}(t)$ by an angle ϕ that increases from 0° (at $f \ll f_c$) to 90° (at $f \gg f_c$), with $\phi = -45^\circ$ exactly at $f = f_c$.

Measurement Procedure

The circuit is set-up on a bread-board. The input voltage $v_{in}(t)$ is given from a **Function Generator**. The control knobs in the front-panel of the function generator can set the amplitude, function (sine, triangle, rectangular, square waveforms), frequency, and DC offset. Before applying the input to the circuit, observe the waveform on a Digital Storage Oscilloscope (DSO) to ensure that the input signal is as per the requirements (amplitude, form, frequency and offset).

Relating Time Delay to Phase Shift

On a DSO, phase shift is most directly obtained from the time delay Δt between corresponding zero crossings of $v_{in}(t)$ and $v_{out}(t)$, together with the period T of the waveform:

$$\phi(\text{in degrees}) = \frac{\Delta t}{T} \times 360^\circ = \Delta t f \times 360^\circ. \quad (5)$$

See the figure:

Care must be taken to note *which* channel leads and which lags, since Equation (5) gives only the magnitude of ϕ unless signs are tracked from the oscilloscope trace.

The Lissajous Method

If the DSO is operated in X-Y mode (CH1 on the horizontal axis, CH2 on the vertical axis), two sinusoids of the same frequency but different phase produce an elliptical trace (a *Lissajous*

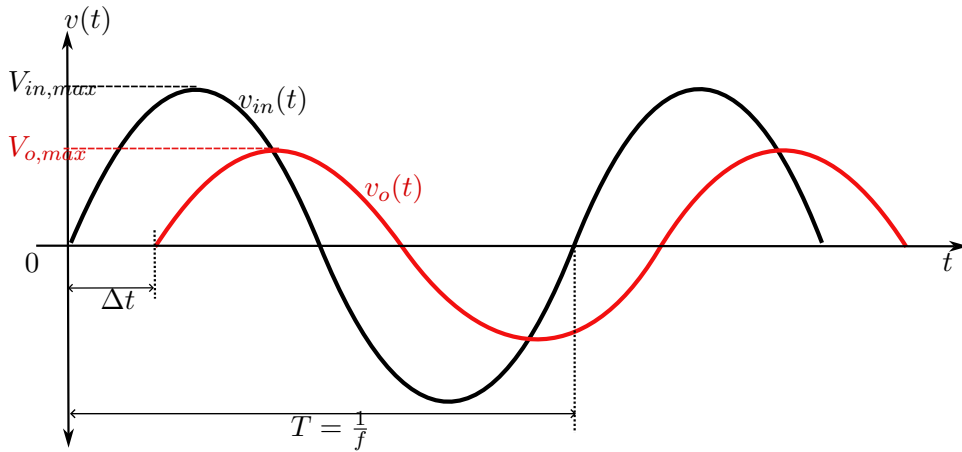


Figure 2: Input and output waveforms and salient parameters

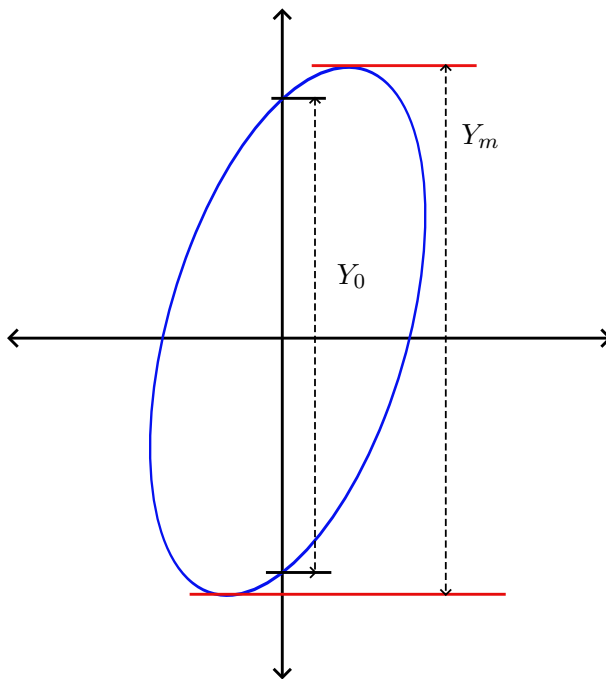


Figure 3: Lissajous pattern and phase-shift calculation

figure)⁴. See figure 3. Y_0 is the distance between the y-intercepts and Y_m is the distance between the extreme vertical points. The phase shift is then given by:

$$\phi = \sin^{-1} \left(\frac{Y_0}{Y_m} \right). \quad (6)$$

This method is a useful independent cross-check of the time-difference method and is more accurate near $\phi = 90^\circ$, where Δt becomes difficult to resolve on the time-domain trace.

Note

Modern DSOs have advanced measurement features that enable direct reading of the phase difference between two waveforms on screen, in addition to cursor measurements. For more details, consult the operating manual of the DSO model.

⁴You may explore [this online](#) simulation to learn more about Lissajous patterns.

Apparatus Required

S.No.	Equipment	Specification
1	Digital Storage Oscilloscope	2-channel, 20–50 MHz bandwidth
2	Function Generator	Sine wave, 1 Hz – 1 MHz
3	Resistor	1 (value assigned in lab, e.g. 1 k Ω)
4	Capacitor	1 (value assigned in lab, e.g. 0.1 μ F)
5	Breadboard and connecting wires	–
6	BNC–BNC and BNC–crocodile probes	As required

Table 7: Equipment list.

Pre-Lab Preparation (to be completed *before* coming to the laboratory)

Given the values of R and C assigned to your batch:

- Calculate the cutoff frequency f_c using Equation (2).
- Using Equations (3) and (4), complete the theoretical prediction table below for input amplitude $V_m = 1$ V (peak), at frequencies referenced to f_c .
- Sketch, on graph paper, the expected shape of the Lissajous figure at $f = 0.1f_c$, $f = f_c$, and $f = 10f_c$.

f	f/f_c	$ H(j\omega) $	$V_{out,pp}$ (theory)	ϕ (theory, deg)	Δt (theory)
$0.1f_c$	0.1				
$0.5f_c$	0.5				
f_c	1				
$2f_c$	2				
$10f_c$	10				

Table 8: Theoretical prediction table (to be filled in during pre-lab preparation).

Experimental Procedure (Overview)

- Connect the series RC network as shown in Figure 1. Set the function generator to produce a sine wave of the specified amplitude and set its frequency to $0.1f_c$. Make sure that the output has no DC offset, i.e., the output is a pure ac sine wave.
- Connect DSO CH1 across the function generator output (v_{in}) and CH2 across the capacitor (v_{out}), ensuring a *common ground* between the generator, the circuit, and both probes.
- Use AUTOSSET (or manual Volts/Div and Time/Div adjustment) to display 2–3 complete cycles of both waveforms on screen.
- Using the DSO's horizontal cursors, measure the peak-to-peak voltage $V_{in,pp}$ and $V_{out,pp}$ for each channel.
- Using the vertical (time) cursors, measure the period T of the waveform and hence the frequency $f = 1/T$; cross-check against the function generator's dial/display setting.

6. Using the time cursors, measure the time delay Δt between corresponding zero crossings of v_{in} and v_{out} and calculate ϕ from Equation (5).
7. Switch the DSO to **X–Y mode** and record the resulting Lissajous figure. Measure Y_0 and Y_m and calculate ϕ from Equation (6).
8. Repeat Steps 4–7 for each frequency listed in Table 8 (and any additional frequencies specified by the instructor).
9. Tabulate all measured quantities alongside the theoretical predictions and calculate the percentage deviation.

Precautions

- Ensure the ground leads of the function generator and both oscilloscope probes are connected to a common reference node to avoid ground-loop errors.
- Compensate the oscilloscope probes (using the built-in calibration signal) before taking measurements. You may use the probe calibration facility of the DSO.
- Keep the function generator output amplitude constant while varying frequency; check it does not change due to output loading.⁵
- Use the $\times 1$ or $\times 10$ probe setting consistently and account for the attenuation factor in the DSO's channel settings.

Practical Observations

Append the table 8 with additional columns of practical observations. Provide your inferences and conclusions/results.

Pre-Lab Questions

- Q1. Derive Equation (1) from first principles using the voltage divider rule in the phasor domain.
- Q2. Show that at $f = f_c$, the output amplitude is $1/\sqrt{2}$ times the input amplitude and the phase shift is exactly -45° .
- Q3. If the output were taken across R instead of C , what type of filter results? Derive the new expressions for $|H(j\omega)|$ and $\phi(f)$.
- Q4. Why is the Lissajous method more reliable than the time-delay method for measuring phase shifts close to 90° ?
- Q5. List two sources of error in measuring Δt directly from the DSO screen and suggest how each may be minimized.

⁵Loading happens when the Circuit Under Test (CUT) has low input impedance compared to the function generator's output impedance.

Pre-Lab 2:

Introduction to Circuit Simulation using LTspice

Objectives

- (i) To install and set up LTspice, a free SPICE-based analogue circuit simulator.
- (ii) To become familiar with the basic workflow of schematic capture, simulation setup, and waveform viewing in LTspice.
- (iii) To simulate a simple RC network and verify the results against hand-calculated values, in preparation for comparing simulation with hardware measurements in later experiments.

What is LTspice?

LTspice is a high-performance SPICE (Simulation Program with Integrated Circuit Emphasis) simulator, schematic capture tool, and waveform viewer, distributed free of charge by Analog Devices. It allows a circuit to be drawn on screen, assigned component values and a source, and then analysed numerically – computing node voltages and branch currents as functions of time (transient analysis), frequency (AC analysis), or DC bias (DC sweep/operating point analysis) – without building any physical hardware. Throughout this course, LTspice will be used *before* each hardware experiment to predict expected waveforms, and the simulated results will be compared with those measured on the DSO in the laboratory.

Basic Workflow

1. **Create a new schematic.** File → New Schematic, then save it with a suitable file-name before adding components (LTspice requires the file to be saved to run a simulation correctly).
2. **Place components.** Use the toolbar icons (or keyboard shortcuts) to place a resistor (R), capacitor (C), inductor (L), diode (D), ground (G), and a voltage source. Rotate/mirror using Ctrl+R/Ctrl+E.
3. **Wire the circuit.** Use F3 (or the wire tool) to connect component terminals. Every circuit *must* include a ground node (node 0), or the simulation will fail.
4. **Set component values.** Right-click a component and enter its value (e.g., 1k, 0.1u). For the source, right-click and choose the appropriate function (e.g., SINE for a sinusoidal source, specifying amplitude, frequency, and DC offset).

5. **Choose the analysis type.** Simulate → Edit Configure Analysis⁶:

- `.tran` – transient (time-domain) analysis, for viewing waveforms vs. time.
- `.ac` – AC sweep, for frequency response (Bode plots).
- `.op` – DC operating point.

6. **Run the simulation.** Simulate → Run (or the RUN/PLAY icon). A waveform viewer window opens automatically.

7. **Probe voltages and currents.** Click on a wire/node in the schematic to display its voltage, or click on a component to display the current through it, in the waveform viewer. To show the voltage between nodes, click and drag the probe between the nodes.

8. **Add cursors and measure.** Use **Ctrl**+left-click on a trace label to attach a cursor; the cursor readout gives precise voltage, time, and (for two cursors) Δt and ΔV – directly analogous to DSO cursor measurements.

Pre-Lab Exercise

Using LTspice, draw the series RC network of Pre-Lab 1 (Figure 1) with the R and C values assigned to your batch:

- Apply a **SINE** source with amplitude 1 V (peak) at $f = f_c$ (calculated in Pre-Lab 1) and run a `.tran` simulation for at least 5 cycles.
- Using cursors, measure $V_{out,pp}$ and the time delay Δt between v_{in} and v_{out} ; hence compute ϕ using Equation (5), and compare with the theoretical value from Table 8.
- Run a `.ac` simulation from $0.01f_c$ to $100f_c$ and plot the magnitude (in dB) and phase of v_{out}/v_{in} . Identify f_c on the plot as the -3 dB point.
- Save screenshots of both simulations for submission along with your pre-lab record.

Pre-Lab Questions

- Q1. What is the difference between `.tran` and `.ac` analysis, and when would each be preferred?
- Q2. Why must every LTspice schematic contain a ground (node 0) reference?
- Q3. What is the purpose of the *maximum timestep* parameter in a `.tran` command, and what artefact can result if it is left too large?
- Q4. How would you model a real (non-ideal) function generator output impedance (typically 50Ω) in an LTspice source?

Resources

- LTspice download and official documentation (Analog Devices):
<https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html>
- LTspice Quick Reference / Getting Started guide: <https://www.analog.com/en/education/education-library/ltspice-getting-started-guides.html>

⁶in Version 24.x

- Analog Devices LTspice video tutorial series (YouTube):
<https://www.youtube.com/@AnalogDevicesInc>
- LTwiki – community-maintained reference for LTspice syntax and models:
<https://ltwiki.org/>

Other free simulation tools

There are other SPICE-based free tools available which may also be used instead of LTspice. Some popular tools are listed here.

1. Qspice. Qspice is created by the creator of LTspice, Mike Engelhardt, as a modern, fast simulator: <https://p.qorvo.com/get-qspice>
2. Ngspice (requires a schematic capture tool): <https://ngspice.sourceforge.io>
3. eSim is an EDA package that goes beyond spice. Maintained by FOSSEE project under IITB. It uses ngspice as the simulation engine, and KiCad schematic for schematic entry.
<https://esim.fossee.in/home/>

Experiment 1

Clipping and Clamping Circuits using Diodes

1.1 Aim

- (i) To study the operation of series and shunt diode clipping circuits and to observe the effect of biasing on the clipping level.
- (ii) To study the operation of positive and negative diode clamping circuits.
- (iii) To design clipper and clamper circuits for specified output requirements.
- (iv) To verify, using a DSO, that the observed waveforms agree with the theoretically predicted transfer characteristics.

1.2 Pre-Lab Exercise

Before performing the experiment, simulate the circuits and verify the designs using LTspice or any other circuit simulation software.

1.3 Equipment/Components Required

S.No.	Equipment/Component	Specification
1	Digital Storage Oscilloscope	2-channel, 20–50 MHz bandwidth
2	Function Generator ¹	Sine wave, 1 Hz – 1 MHz
3	Regulated DC Power Supply	0–30 V
4	Silicon diode	1N4007 (or equivalent), 2 nos.
5	Zener diodes	$V_z = 5.6$ V, 2 nos.
6	Resistors	1 k Ω – 100 k Ω
7	Capacitors	1 μ F – 10 μ F
8	Breadboard and connecting wires	–

Table 1.1: Equipment and component list.

1.4 Theory

Diode clipping and clamping circuits exploit the nonlinear, near-ideal-switch behaviour of a p – n junction diode to reshape a waveform. A *clipper* removes (clips) a portion of the input waveform

above and/or below a reference level, while a *clammer* shifts the entire waveform up or down without altering its shape, thereby changing its DC reference level.

Throughout this document, the diode is modelled as an ideal switch in series with a cut-in voltage $V_F \approx 0.7\text{ V}$ (silicon) and a forward resistance $R_f \approx 0$; the reverse resistance is assumed infinite.

1.4.1 Clipping Circuits

(a) Series (Half-Wave) Clipper. ² Figure 1.1 shows a series diode clipper. The diode conducts only when v_{in} forward-biases it, so the circuit passes only one half-cycle of the input to the output. The output voltage across the resistance R_L will be $v_{in}(t) - V_F$, for $v(t) > 0$, and 0 V for $v(t) < 0$.

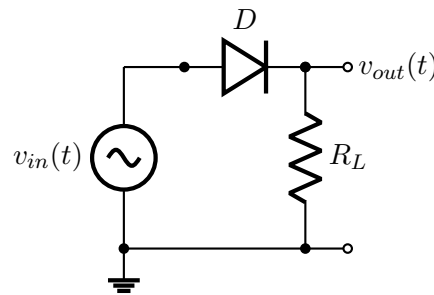


Figure 1.1: Series (half-wave) diode clipper. The diode conducts on the positive half-cycle and passes it to R_L ; the negative half-cycle is clipped.

(b) Shunt (Parallel) Clipper – Positive Clipper. Figure 1.2a shows a shunt clipper with the diode across the output. Whenever $v_{in} > V_F (\approx 0.7\text{ V})$, the diode conducts and clips the output at V_F . For $v_{in}(t) < V_F$, the diode is OFF and $v_{out} = v_{in} \frac{R_L}{R_L + R}$. If $R_L \gg R$, $v_{out} \approx v_{in}$, during the negative cycle. The waveforms expected are as given in Fig. 1.2c.

The positive portion of the input waveform is clipped off, hence the name positive clipper.

(c) Shunt (Parallel) Clipper – Negative Clipper. Figure 1.3a shows a shunt negative clipper with the diode across the output, placed in the opposite polarity compared to the positive clipper. Whenever $v_{in}(t) < -V_F$, the diode conducts and clamps the output near $-V_F$; for $v_{in}(t) > -V_F$, the diode is OFF and $v_{out} = v_{in} \frac{R_L}{R_L + R}$. If $R_L \gg R$, $v_{out} \approx v_{in}$, during the positive cycle. The waveforms expected are as given in Fig. 1.3c.

The negative portion of the input waveform is clipped off, hence the name negative clipper.

(d) Biased Clippers. A DC bias source V_R in series with the diode (Figure 1.4) shifts the clipping level from V_F to $(V_R + V_F)$ or $(V_R - V_F)$, depending on polarity. This allows the clipping level to be set anywhere on the waveform. There are two ways to achieve the biasing. One is to provide a direct dc source. The other is to use a Zener diode or series combination of diodes to achieve the bias level.

Fig. 1.4a shows a biased positive clipper with a dc bias. Fig. 1.4b shows the implementation with a Zener diode. The clipping levels are $V_R + V_F$ and $V_Z + V_F$ respectively. Fig. 1.5a shows the sample waveforms.

In a similar way, a **Biased Negative Clipper** can be implemented by reversing the diode polarity. The figures 1.7a and 1.7b show the circuit implementations.

The sample waveforms for the biased negative clipper are as shown in Fig. 1.8a.

²This circuit need not be assembled and tested. This is presented here for completeness.

Double (Two-Level) Clipper or Slicer. Two diodes, oppositely poled and biased at V_{R1} and V_{R2} , in parallel (Figure 1.10a), clip both the positive peak above $(V_{R1} + V_F)$ and the negative peak below $-(V_{R2} + V_F)$, producing a trapezoidal output for a sinusoidal input of sufficient amplitude. If the amplitude of the sine wave is much larger compared to the clipping level, the output waveform approximates a square-wave. This circuit is also known as a ‘slicer’.

Instead of DC biasing voltages, zener diodes can be used.

Transfer Characteristics

The transfer characteristics are graphs which plot the output (voltage) against the input (voltage). They give a visual indication of about the relationship between the input and the output. They also visualise which region of the operation of the circuit is linear. The transfer characteristics of the various clipper circuits are shown along with their respective circuit diagrams, in the section 1.6.1.

1.4.2 Clamping Circuits

A clamper consists of a diode, a capacitor, and a resistor, arranged so that the capacitor charges (through the low forward resistance of the diode) to the peak of the input during one half-cycle and holds this charge (discharging slowly through R_L , with $R_L C \gg T$) during the rest of the cycle, thereby shifting the DC level of the waveform. It essentially ‘clamps’ the input signal up or down by the peak voltage. It is also called a DC restorer.

Mathematically, for an input signal given by $v_{in}(t) = V_m \sin(\omega t)$, the clamper output is $v_o(t) = \pm V_m + V_m \sin(\omega t)$.

(a) Negative Clamper. Figure 1.11a shows a negative clamper, which shifts the waveform downward so that its positive peak is clamped at (approximately) 0 V (or at V_F , accounting for diode drop).

(b) Positive Clamper. Reversing the diode polarity (Figure 1.12a) clamps the negative peak of the waveform at approximately 0 V, shifting the entire waveform upward.

For an input sine wave of amplitude V_m , the steady-state clamped output ideally swings between:

$$\text{Negative clamper:} \quad v_{out} \in [-2V_m + V_F, V_F], \quad (1.1)$$

$$\text{Positive clamper:} \quad v_{out} \in [-V_F, 2V_m - V_F]. \quad (1.2)$$

The time constant $R_L C$ must satisfy $R_L C \gg T$ (the input period) so that the capacitor voltage does not decay appreciably between successive charging peaks (poor choice of $R_L C$ leads to *waveform tilt/sag*).

1.5 Design Example

- Select the operating signal’s frequency and amplitude, $V_m = 10$ V and $f = 100$ Hz.
- The series resistor R should not be too large or too small. R_{min} is decided by the maximum current in the circuit. A typical value for R would be 1 k Ω . Since $R_L \gg R$, $R_L \geq 100$ k Ω ³. The output voltage peak then would be:

$$V_{out,m} = \frac{V_m R_L}{R + R_L} = \frac{10 \times 100 \text{ k}}{1 \text{ k} + 100 \text{ k}} = 9.9 \text{ V}$$

³For viewing on the DSO, R_L may be removed, then V_{out} would be same as V_{in} , since the input impedance of DSO is in M Ω . In practical circuits where the clipped waveform is applied as input to other circuits, R_L represents the input impedance of those circuits.

- (c) For a **biased clipper** required to clip at a level $V_{clip} < V_m$, select $V_R = V_{clip} - V_F$ using a potential divider from the DC supply, or a battery/regulated source directly. A Zener diode may also be chosen, as described earlier. Let $V_{clip} = 6.3$ V. Then $V_R = V_{clip} - V_F = 6.3 - 0.7 = 5.6$ V. In this case, either a dc voltage of 5.6 V or a Zener diode of $V_Z = 5.6$ V may be used as bias V_R .
- (d) For a **clamper**, choose C such that $R_L C \geq 10T$, where $T = 1/f$ is the period of the lowest input frequency to be used, so that the sag in output during discharge is limited to within an acceptable fraction (typically $< 5\%$) of V_m . For $f = 100$ Hz, $T = 10$ ms. Then, for $C = 1$ μF electrolytic:

$$R_L C \geq 10T \Rightarrow R_L \geq \frac{10 \times 10 \times 10^{-3}}{1 \times 10^{-6}}$$

which yields:

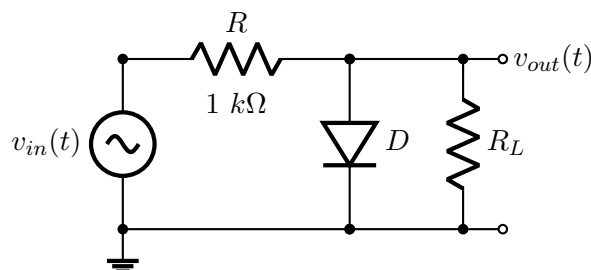
$$R_L \geq 100 \text{ k}\Omega$$

- (e) **Selection of diodes:** For input frequencies less than 1 kHz, line frequency diodes such as 1N4001 may be used. For higher frequencies, these slow diodes are to be replaced with faster signal diodes such as 1N4148.

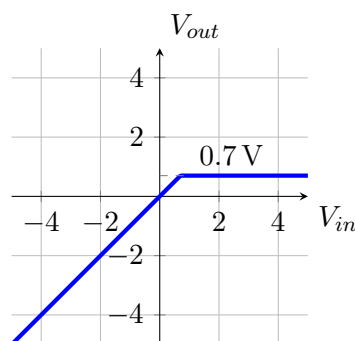
1.6 Circuit Diagrams

1.6.1 Clipping Circuits

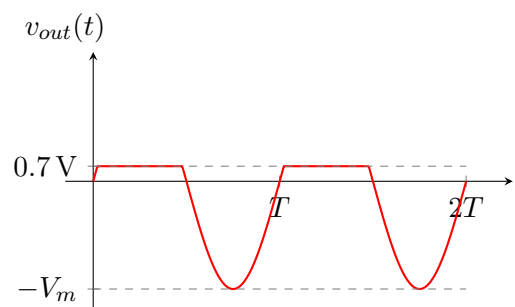
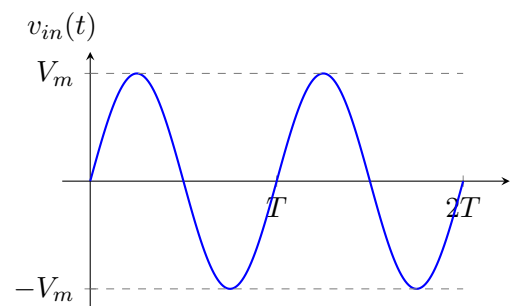
Shunt Positive Clipper



(a) Shunt positive clipper Circuit



(b) Transfer Characteristics



(c) Positive clipper waveforms

Figure 1.2: Positive clipper circuit, input/output waveforms and transfer characteristics

Shunt Negative Clipper

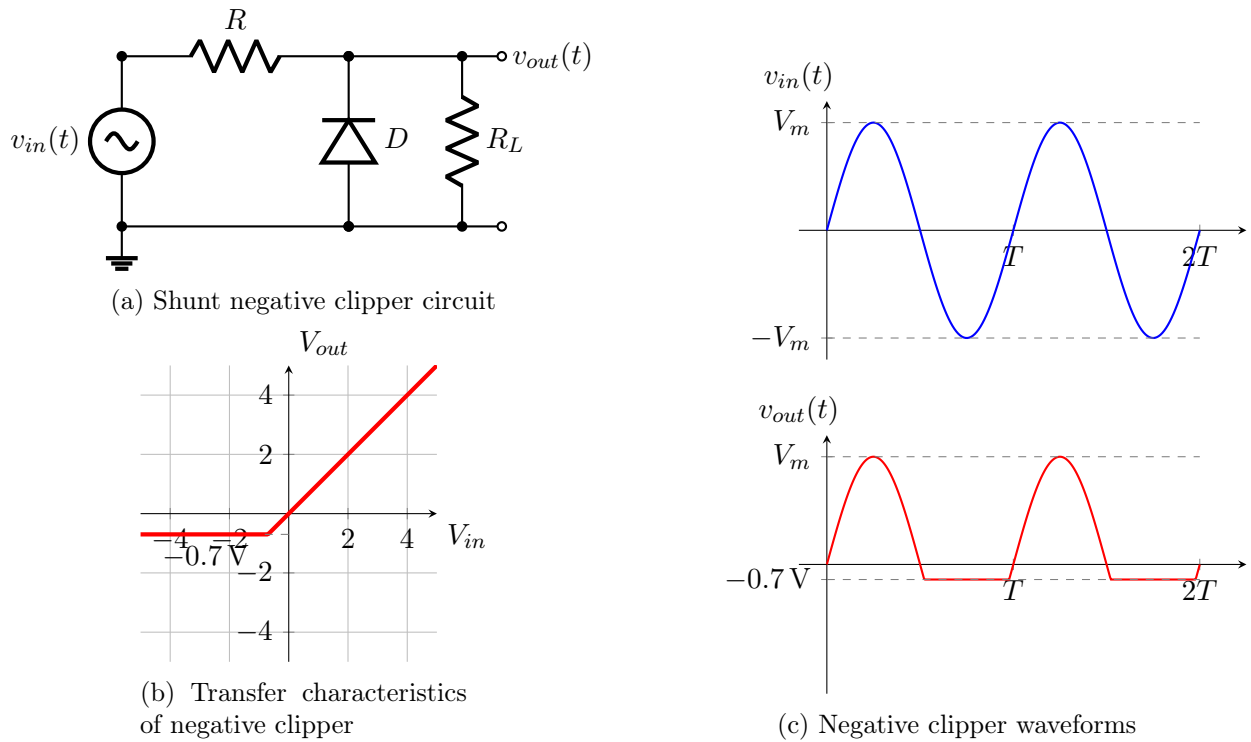


Figure 1.3: Diode clipper circuit with input and output waveforms

Biased Positive Clippers

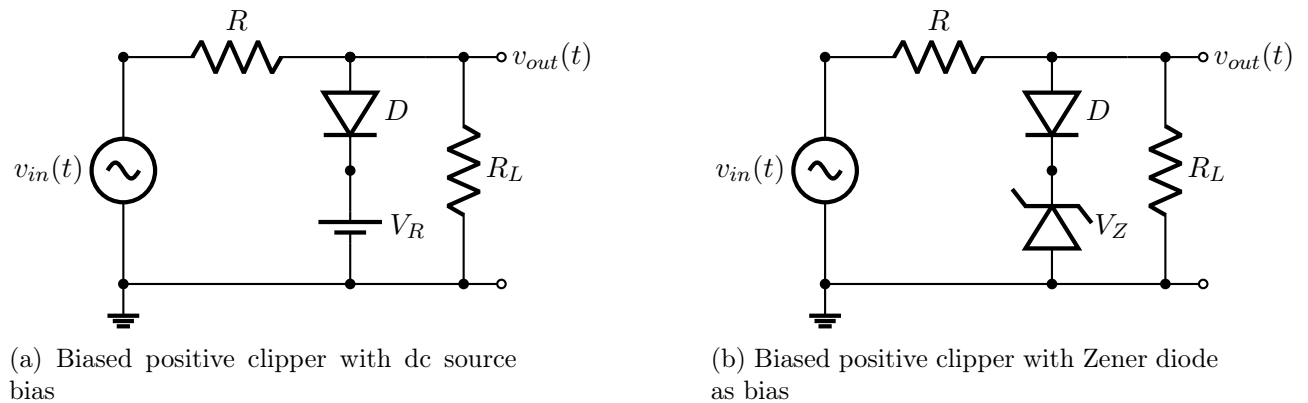


Figure 1.4: Biased positive clipper configurations

The clipping level is moved to negative, by reversing the bias voltage V_R in Fig. 1.4a. However, the same effect cannot be achieved through the circuit in Fig. 1.4b by reversing the Zener.

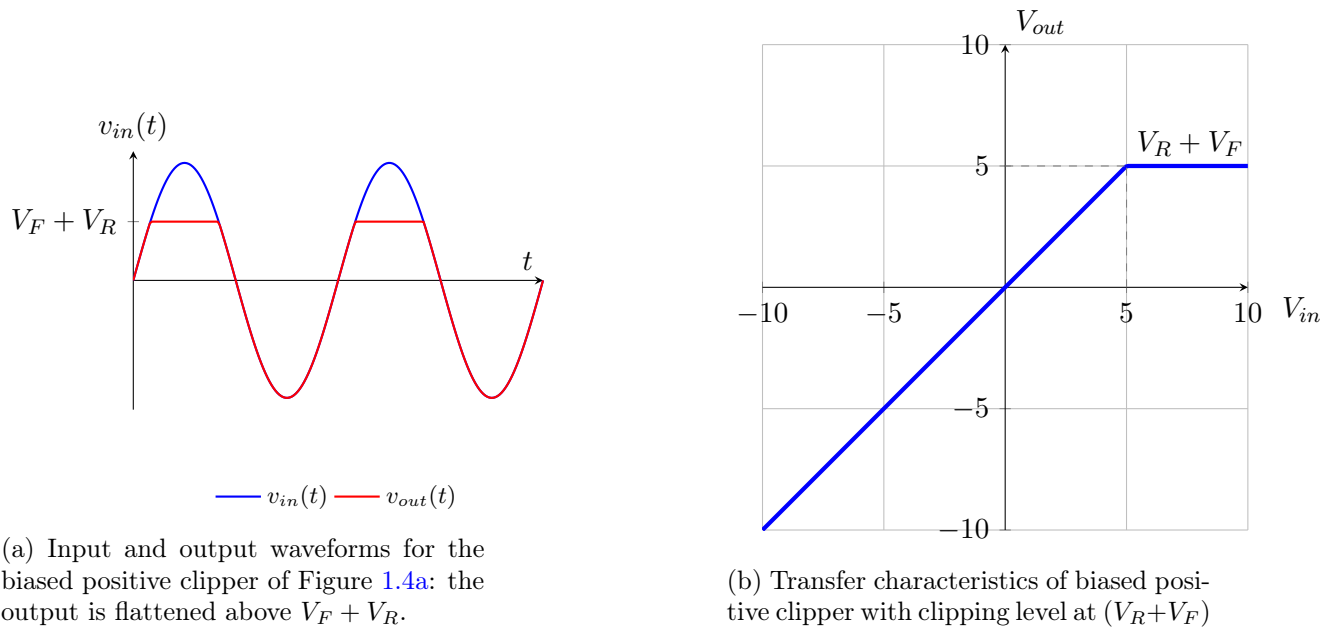


Figure 1.5: Biased positive clipper waveforms and transfer characteristics for a positive clipping level

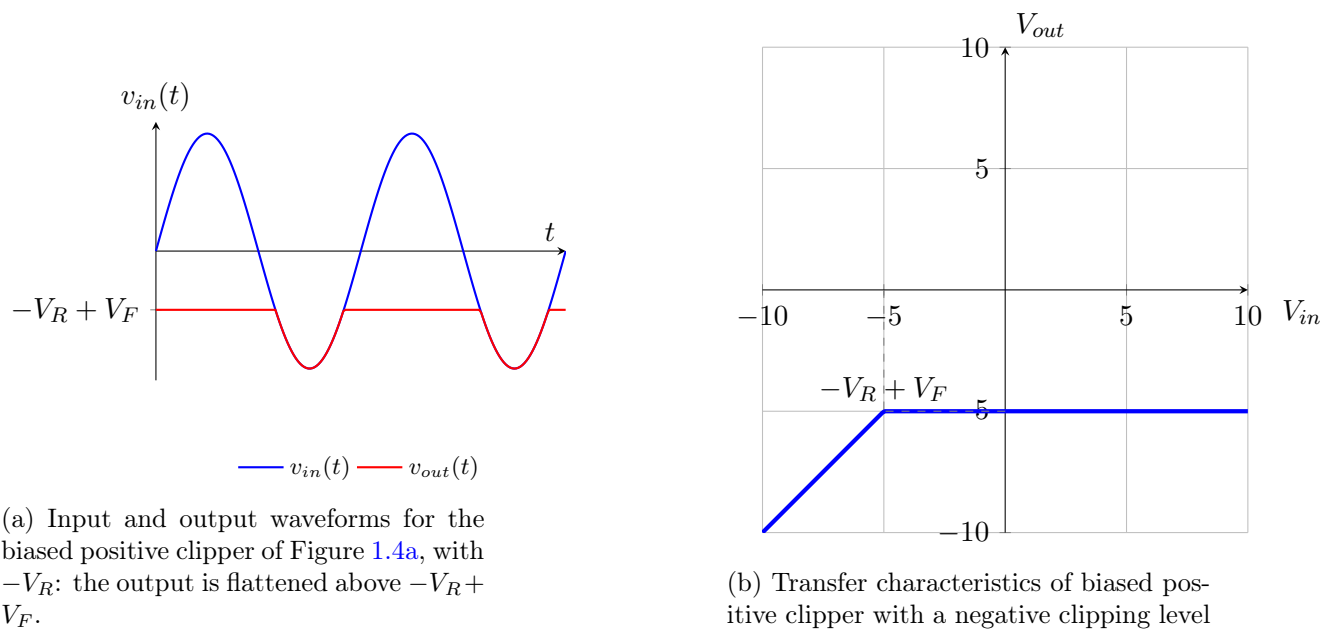
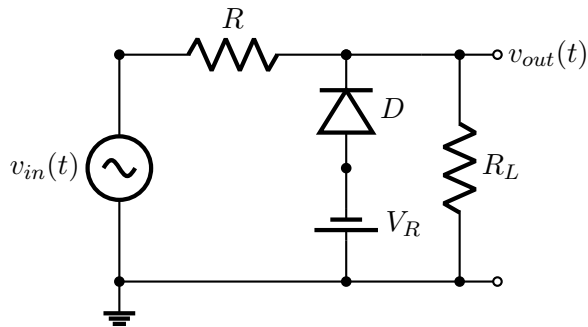
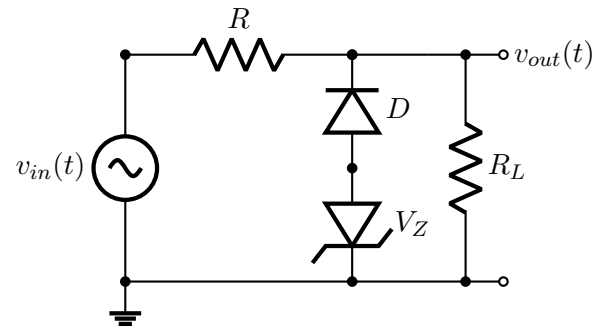


Figure 1.6: Waveforms and transfer characteristics of biased positive clipper with a negative clipping level

Biased Negative Clippers

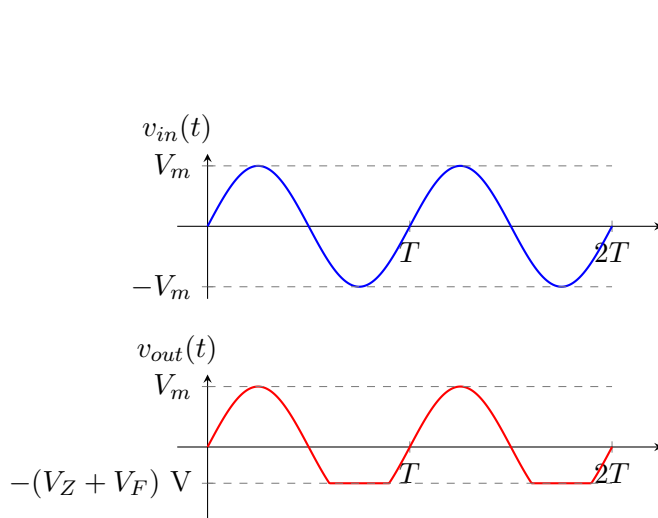


(a) Biased negative clipper with dc source bias.

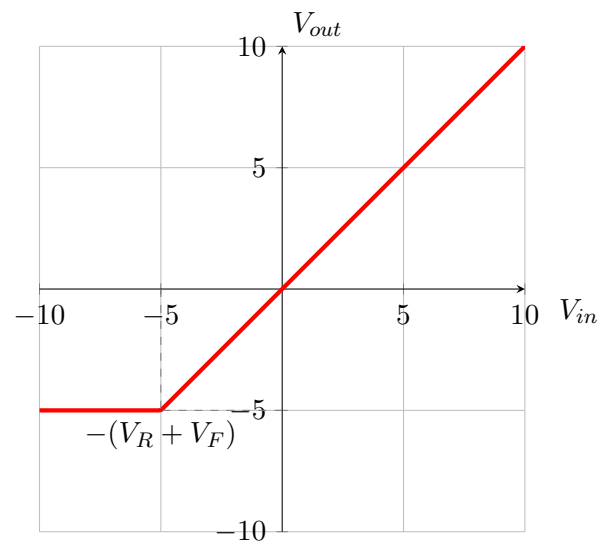


(b) Biased negative clipper with Zener diode for biasing.

Figure 1.7: Biased Negative Clipper Implementations

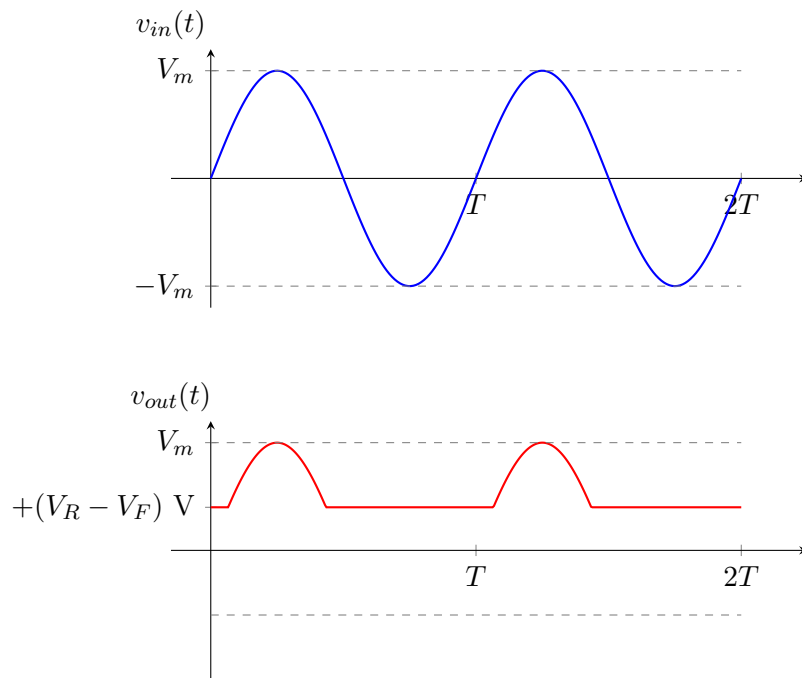


(a) Biased negative clipper waveforms with negative clipping level (For Zener based biasing)



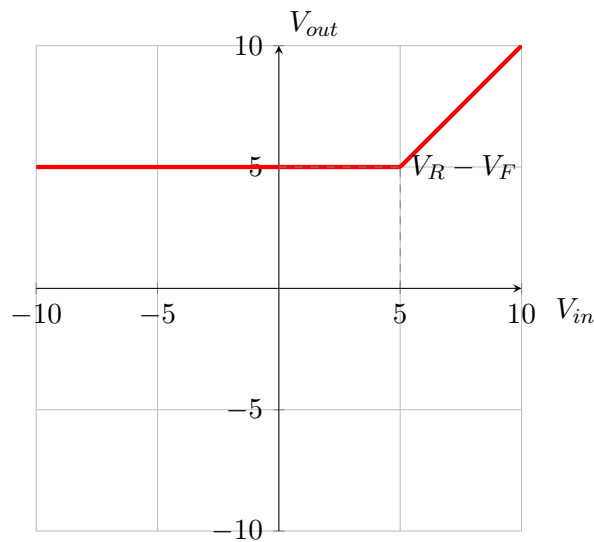
(b) Transfer characteristics of biased negative clippers

Figure 1.8: Biased negative clipper: Waveforms and transfer characteristics



(a) Biased negative clipper waveforms with positive clipping level (polarity of V_R reversed in Fig. 1.7a)^a

^aReversal of Zener polarity in Fig. 1.7b to achieve the same results is not possible. Can you think why?



(b) Transfer characteristics of biased negative clippers: clipping at positive level

Figure 1.9: Biased negative clipper with positive level: Waveforms and transfer characteristics

Slicer/Double Clipper

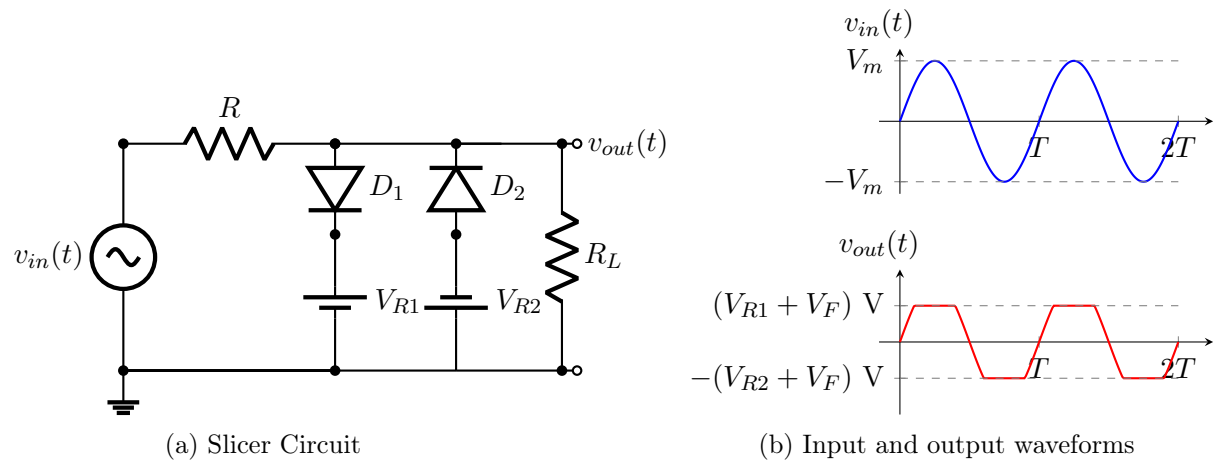


Figure 1.10: Slicer circuit and waveforms

1.6.2 Clamping Circuits

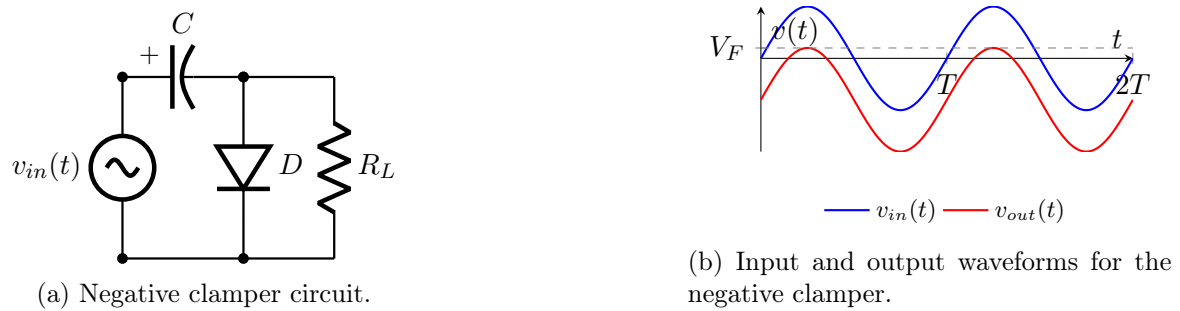


Figure 1.11: Negative Clamper: Circuit and waveforms

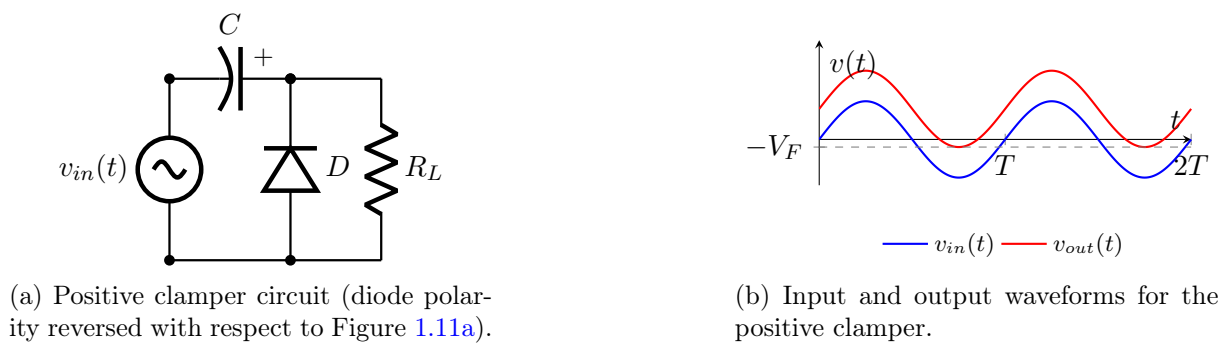


Figure 1.12: Positive clamper: Circuit and waveforms

Note

While preparing the lab-report, circuit diagrams shall contain proper labels for the components, including the component part numbers (such as 1N4001), and values/parameters (example: $10 V_{pp}$, Sine, 100 Hz, for the source; R_1 , 1 k Ω , etc.).

1.7 Experiment Procedure

1. Assemble the shunt positive clipper of Figure 1.2a on the breadboard. Set the function generator to a sine wave of frequency 100 Hz and amplitude $10 V_{pp}$.
2. Connect DSO CH1 to v_{in} and CH2 to v_{out} , with a common ground. Display both waveforms simultaneously and sketch/record them.
3. Replace the shunt positive clipper with the shunt negative clipper. (Figure 1.3a). Record v_{in} and v_{out} , and measure the clipping level from the DSO screen using the horizontal cursors.
4. Repeat for the biased clipper (Figure ??) for at least two values of V_R , and the double clipper (Figure 1.10a). *(Note that the peak of the selected input waveform is to be higher than the clipping level for the case of biased clippers).*
5. For each clipper configuration, tabulate the theoretically expected and experimentally measured clipping levels.
6. Assemble the negative clamper (Figure 1.11a) using the design values of R_L and C computed in the Design Procedure. Apply the same sine wave and record v_{in} and v_{out} on the DSO, noting the DC shift and any tilt/sag in the output.
7. Repeat for the positive clamper (Figure 1.12a).
8. For each clamper, measure $V_{out,max}$ and $V_{out,min}$ using DSO cursors and compare with Equations (1.1)–(1.2).
9. Vary the input frequency (e.g., to 100 Hz) while keeping $R_L C$ fixed, and observe/record the effect on clamper output tilt. Explain the observation.

1.8 Precautions

- Observe correct diode polarity while assembling each circuit; an inverted diode produces the mirror-image (and often misleading) result.
- Do not exceed the diode's maximum forward current or reverse voltage ratings (Important if you are using a step-down transformer to give the input signal (50 Hz), instead of a function generator).
- Ensure a common ground between the function generator and both DSO channels.
- Electrolytic capacitors, if used, connect with correct polarity as per the expected DC level at that node.

1.9 Expected Waveforms

The sample waveforms for the clipping circuits are as shown in Figs. 1.2c, 1.3c, 1.5a, 1.8a and 1.10b. The waveforms for the clamping circuits are as shown in Figs. 1.11b, 1.12b. When recording waveforms, note down all the salient points/features of the waveforms, such as voltage/time details.

1.10 Observation Tables

The observation table for the clipping circuits is given in Table 1.2. Use Table 1.3 to tabulate the Clamper circuit observations.

Circuit	V_{clip} (theory)	V_{clip} (measured)	% Error
Shunt positive clipper			
Shunt negative clipper			
Biased positive clipper (V_{R1})			
Biased negative clipper (V_Z)			
Double clipper (upper level)			
Double clipper (lower level)			

Table 1.2: Clipping circuits: theoretical vs. measured clipping levels.

Circuit	$V_{out,max}$ (theory)	$V_{out,max}$ (meas.)	$V_{out,min}$ (theory)	$V_{out,min}$ (meas.)
Negative clamper				
Positive clamper				

Table 1.3: Clamping circuits: theoretical vs. measured output extremes.

1.11 Results

1. The diode-based clipping circuits with and without external bias are set-up and the output waveforms were studied.
2. The diode-based clamping circuits were set-up and the waveforms were studied.

Explore further

Experiment with higher frequency signals. What do you observe? Can you explain why circuit performance is not as expected at higher frequencies?

1.12 Questions

- Q1. Distinguish between series and shunt clipper configurations in terms of output impedance and loading effects.
- Q2. Derive the expression for the clipping level of the biased clipper of Figure ?? in terms of V_R and V_F .
- Q3. Why must the time constant $R_L C$ of a clamper be much greater than the period of the input waveform? What happens if $R_L C$ is comparable to T ?
- Q4. Sketch the output waveform of the double clipper (Figure 1.10a) for an input triangular wave of amplitude greater than $(V_{R1} + V_{R2} + 2V_F)$.
- Q5. A clamper is required to shift a $10 V_{pp}$, 1 kHz sine wave so that its output never goes negative. Design suitable values of R_L and C , assuming a maximum permissible sag of 2%.
- Q6. Explain why a clipper is sometimes called a *limiter* and give one practical application each of a clipper and a clamper in electronic systems.

- Q7. Why do clipper circuits do not perform as intended when diodes like 1N4001 are used, at higher frequencies, such as 50 kHz?

Experiment 2

Zener Diode Voltage Regulator

2.1 Aim

To design, and construct a Zener diode based DC voltage regulator for a given load current and input voltage range, and to study its line regulation and load regulation characteristics.

2.2 Pre-Lab Exercise

Before performing the experiment, simulate the circuits and verify the designs using LTspice or any other circuit simulation software.

2.3 Apparatus / Components Required

S.No.	Component	Specification
1	Zener diode	$V_Z = 5.6 \text{ V}$, $P_{Z(\max)} = 400 \text{ mW}$
2	Resistor R_S	As designed (series/current-limiting resistor)
3	Resistor (as R_L)	1–10 k Ω
4	Regulated DC Power Supply	0–15 V, variable
5	Digital Multimeter	–
6	Breadboard and connecting wires	–

2.4 Theory

A Zener diode, when reverse biased beyond its breakdown voltage V_Z , maintains a nearly constant voltage across its terminals over a wide range of reverse current. This property is exploited to build a simple shunt voltage regulator: a series resistor R_S limits the current drawn from the unregulated input V_{in} , while the Zener diode, connected in parallel with the load R_L , clamps the output voltage to V_Z as long as it is operated between its minimum breakdown-sustaining current $I_{Z(\min)}$ and its maximum rated current $I_{Z(\max)}$ (set by its power rating).

The regulator must continue to hold $V_{out} \approx V_Z$ under two independent disturbances:

- **Line regulation:** V_{in} varies over its specified range while the load current I_L is held constant.
- **Load regulation:** The load current I_L varies (via R_L) while V_{in} is held constant.

2.5 Design

Given specifications.

$$\begin{aligned} V_Z &= 5.6 \text{ V} \\ I_{L(\max)} &= 5 \text{ mA} \\ V_{in} &: 8 \text{ V to } 15 \text{ V} \quad (V_{in(\min)} = 8 \text{ V}, V_{in(\max)} = 15 \text{ V}) \\ P_{Z(\max)} &= 400 \text{ mW} \end{aligned}$$

Step 1: Maximum Zener current from the power rating.

$$I_{Z(\max)} = \frac{P_{Z(\max)}}{V_Z} = \frac{400 \text{ mW}}{5.6 \text{ V}} \approx 71.4 \text{ mA}$$

Although this is the maximum Zener current, to operate without damage, a much lower current is assumed. Hence, an $I_{Z(max)}$ is assumed to be 50 mA.

Step 2: Choose minimum Zener current. To ensure the diode stays in the breakdown region even at worst case (minimum V_{in} , maximum I_L), a minimum holding current is chosen as

$$I_{Z(\min)} = 5 \text{ mA} \quad (\text{a conservative, commonly used design margin})$$

Step 3: Design R_S for the worst case (minimum V_{in} , maximum I_L). At $V_{in(\min)}$, the source must supply both $I_{L(\max)}$ and $I_{Z(\min)}$:

$$R_S = \frac{V_{in(\min)} - V_Z}{I_{L(\max)} + I_{Z(\min)}} = \frac{8 - 5.6}{(5 + 5) \times 10^{-3}} = \frac{2.4}{0.010} = 240 \, \Omega$$

Choosing the nearest standard (E12) value: $R_S = 220 \, \Omega$.

Step 4: Verify at minimum V_{in} , full load (checks regulation is maintained).

$$I_T = \frac{V_{in(\min)} - V_Z}{R_S} = \frac{2.4}{220} \approx 10.9 \text{ mA}$$

$$I_Z = I_T - I_{L(\max)} = 10.9 - 5 = 5.9 \text{ mA} \quad (> I_{Z(\min)} = 5 \text{ mA}) \quad \checkmark$$

Step 5: Verify at maximum V_{in} , no load (checks Zener power rating is not exceeded).

$$I_Z = \frac{V_{in(\max)} - V_Z}{R_S} = \frac{15 - 5.6}{220} \approx 42.7 \text{ mA}$$

$$P_Z = I_Z \times V_Z = 42.7 \text{ mA} \times 5.6 \text{ V} \approx 239 \text{ mW} \quad (< 400 \text{ mW}) \quad \checkmark$$

Step 6: Power rating of R_S .

$$P_{R_S} = \frac{(V_{in(\max)} - V_Z)^2}{R_S} = \frac{(9.4)^2}{220} \approx 0.40 \text{ W}$$

Use a $220 \, \Omega$, 1 W resistor (adequate safety margin over the calculated 0.40 W).

Step 7: Selecting Load Resistance R_L . The load resistance is a variable resistance, preferably a rheostat. The load current should be variable the range $0 - I_{Lmax}$. Hence, the minimum resistance should be at least:

$$R_{L,min} \frac{V_Z}{I_{Lmax}} = \frac{5.6 \text{ V}}{5 \text{ mA}} = 1.12 \text{ k}\Omega$$

Hence, choose a $1 \text{ k}\Omega$ resistance as the fixed load, in series with a larger variable resistance ($10 \text{ k}\Omega$) to vary the load current.

Final design summary.

Parameter	Value
Zener diode	5.6 V, 400 mW
Series resistor R_S	220Ω , 1 W
Load current range	0 to 5 mA
Input voltage range	8 V to 15 V
Regulated output	$\approx 5.6 \text{ V}$

2.6 Circuit Diagram

The circuit diagram to be set up is given in Fig. 2.1.

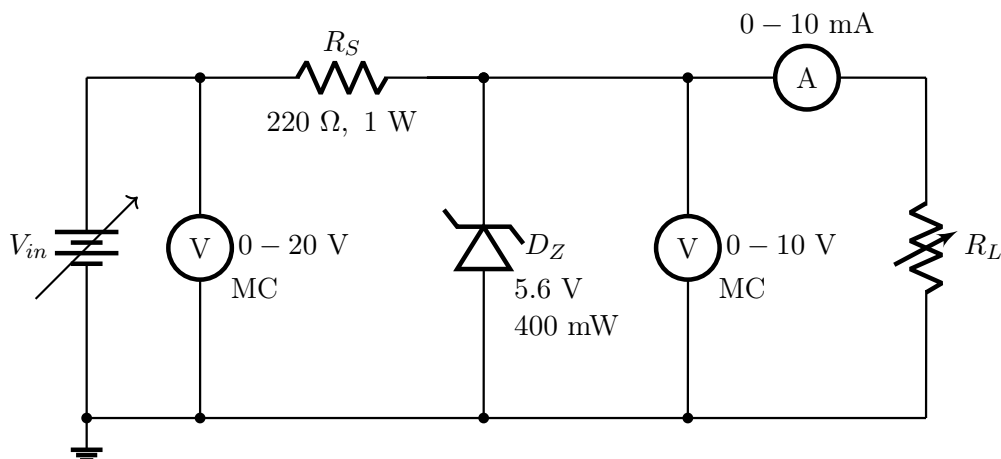


Figure 2.1: Zener diode voltage regulator circuit

2.7 Procedure

(a) Line regulation.

1. Connect the circuit as shown in Figure 2.1 with R_L set to give $I_L = I_{L(max)} = 5 \text{ mA}$ (i.e., $R_L = V_Z / I_{L(max)} \approx 1.12 \text{ k}\Omega$). A rheostat of $2.2 \text{ k}\Omega$ may be used as the load resistance R_L .
2. Vary V_{in} in steps from 8 V to 15 V, keeping R_L fixed.
3. Note V_{out} across the Zener/load for each value of V_{in} .
4. Plot V_{out} versus V_{in} .

(b) Load regulation.

1. Fix V_{in} at a convenient value within the design range (e.g., $V_{in} = 12\text{ V}$).
2. Vary R_L (using the rheostat) so that I_L varies from 0 (open circuit) to $I_{L(\max)} = 5\text{ mA}$.
3. Note V_{out} for each value of I_L .
4. Plot V_{out} versus I_L .

2.8 Tabulation

V_{in} (V)	V_{out} (V)
8	
9	
10	
11	
12	
13	
14	
15	

Table 2.1: Line regulation (I_L held at 5 mA)

I_L (mA)	V_{out} (V)
0	
1	
2	
3	
4	
5	

Table 2.2: Load regulation (V_{in} held constant)**2.9 Model Graph**

Plot two graphs from the observations:

1. V_{out} (Y-axis) vs. V_{in} (X-axis) — should be nearly flat once V_{in} exceeds the value needed to bring the Zener into breakdown, confirming line regulation.
2. V_{out} (Y-axis) vs. I_L (X-axis) — should remain close to V_Z up to $I_{L(\max)}$, with a slight droop as I_L approaches the design limit, confirming load regulation.

2.10 Calculations from Observations

$$\% \text{ Line regulation} = \frac{\Delta V_{out}}{\Delta V_{in}} \times 100\%$$

$$\% \text{ Load regulation} = \frac{V_{out(\text{no load})} - V_{out(\text{full load})}}{V_{out(\text{full load})}} \times 100\%$$

2.11 Result

A Zener diode voltage regulator was designed for $V_Z = 5.6\text{ V}$, $I_{L(\max)} = 5\text{ mA}$, over an input range of 8–15 V, using a 400 mW Zener diode and a series resistor $R_S = 220\ \Omega$. The circuit was constructed and its line regulation and load regulation characteristics were verified experimentally.

The maximum load regulation observed = _____, at $I_L =$ _____

The maximum line-regulation observed = _____, at $V_{in} =$ _____

2.12 Questions

1. Why is the Zener diode operated in reverse breakdown for voltage regulation?
2. What determines the maximum and minimum permissible Zener current?
3. Why must R_S be designed using the minimum input voltage and maximum load current condition?
4. What happens to the regulator if R_L is accidentally open-circuited?
5. Define line regulation and load regulation. What is an ideal value for each?
6. How does the power rating of the Zener diode limit the input voltage range for a fixed R_S ?

Experiment 3

Basic RC Circuits - High-pass and Low-pass Filters

3.1 Aim

To design, construct and study the frequency response (Bode plot) and time-domain behaviour of a first-order RC low-pass filter and a first-order RC high-pass filter for a specified cutoff frequency.

3.2 Pre-Lab Exercise

Before performing the experiment, simulate the circuits and verify the designs using LTspice or any other circuit simulation software.

3.3 Apparatus / Components Required

S.No.	Component	Specification
1	Resistor R	As designed
2	Capacitor C	As designed
3	Function Generator	Sine and square wave, up to 100 kHz
4	Cathode Ray Oscilloscope (CRO)	Dual channel
5	Breadboard and connecting wires	–

3.4 Theory

We have seen the RC circuit in the pre-lab activity, where we used this circuit to familiarise the DSO and the function generator and the measurement techniques.

A first-order RC low-pass filter (LPF) passes low frequencies and attenuates (reduces the amplitude of) high frequencies, while a first-order RC high-pass filter (HPF) does the opposite. Both are built from the same two components, only the position of R and C is interchanged, and both share the same cutoff (or corner) frequency

$$f_c = \frac{1}{2\pi RC}.$$

The attenuation/gain is measured in terms of the unit decibel (dB), where gain in dB is defined as follows:

$$dB = 20 \log_{10} \frac{V_{out}}{V_{in}}$$

Low-pass filter. With R in series and C shunting the output to ground, the transfer function is

$$H_{LP}(f) = \frac{V_{out}}{V_{in}} = \frac{1}{1 + j(f/f_c)}, \quad |H_{LP}(f)| = \frac{1}{\sqrt{1 + (f/f_c)^2}}, \quad \angle H_{LP}(f) = -\tan^{-1}\left(\frac{f}{f_c}\right).$$

High-pass filter. With C in series and R shunting the output to ground, the transfer function is

$$H_{HP}(f) = \frac{V_{out}}{V_{in}} = \frac{j(f/f_c)}{1 + j(f/f_c)}, \quad |H_{HP}(f)| = \frac{f/f_c}{\sqrt{1 + (f/f_c)^2}}, \quad \angle H_{HP}(f) = 90^\circ - \tan^{-1}\left(\frac{f}{f_c}\right).$$

At $f = f_c$, both filters attenuate the signal to $1/\sqrt{2}$ (-3 dB) of the input, with a phase shift of -45° (LPF) or $+45^\circ$ (HPF).

Time-domain view. When the input is a square wave:

- If $RC \gg T$ (period), the LPF cannot follow the fast transitions and behaves like an *integrator*, producing a triangular/rounded output.
- If $RC \ll T$, the HPF passes only the transitions and behaves like a *differentiator*, producing narrow decaying spikes at each edge.

3.5 Design

Specifications.

$$f_c = 1 \text{ kHz}$$

Step 1: Choose a standard capacitor value.

$$C = 0.1 \text{ } \mu\text{F} \quad (\text{standard value})$$

Step 2: Compute R for the required cutoff frequency.

$$R = \frac{1}{2\pi f_c C} = \frac{1}{2\pi \times 1000 \times 0.1 \times 10^{-6}} \approx 1591.5 \text{ } \Omega$$

Nearest standard value: $R = 1.5 \text{ k}\Omega$.

Step 3: Verify actual cutoff frequency.

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 1500 \times 0.1 \times 10^{-6}} \approx 1.061 \times 10^3 \text{ Hz} \approx 1 \text{ kHz} \quad (\text{OK})$$

Final design summary (common to both filters).

Parameter	Value
Resistor R	1.5 k Ω
Capacitor C	0.1 μF
Cutoff frequency f_c	$\approx 1 \text{ kHz}$
Attenuation at f_c	-3 dB ($0.707 \times V_{in}$)
Phase shift at f_c	-45° (LPF), $+45^\circ$ (HPF)

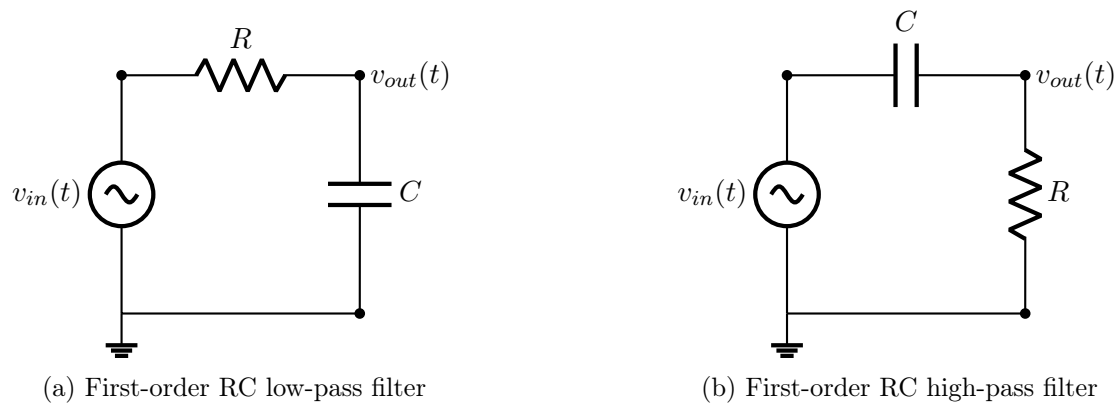


Figure 3.1: First order RC filter circuits

3.6 Circuit Diagrams

The circuits of the RC low-pass and high-pass filters are shown in Fig. 3.1a and Fig. 3.1b respectively.

3.7 Procedure

(a) Frequency response (Bode plot).

1. Connect the circuit as shown in Figure 3.1a (repeat later for Figure 3.1b).
2. Apply a sine wave of constant amplitude V_{in} from the function generator.
3. Vary the frequency from about 10 Hz to 100 kHz in suitable steps (more closely spaced near f_c).
4. At each frequency, measure V_{out} and the phase difference between $v_{in}(t)$ and $v_{out}(t)$ using the CRO (dual trace / Lissajous method).
5. Compute gain in dB as $20 \log_{10}(V_{out}/V_{in})$. Tabulate the observations/calculations as in Table ??.
6. Plot gain (dB) and phase (degrees) versus frequency (log scale) — the Bode magnitude and phase plots. Use a semi-log graph-sheet for plotting the frequency response.
7. Gain (dB) versus frequency for the LPF and HPF, each expected to approach the shapes shown in Figures 3.2 and 3.3, with a -3 dB point at f_c and asymptotic slopes of ± 20 dB/decade away from f_c .
8. Phase (degrees) versus frequency for the LPF and HPF, passing through -45° and $+45^\circ$ respectively at f_c .

(b) Time-domain response.

1. Apply a square wave at a frequency well below f_c (for the LPF, to observe near-integrator behaviour) and well above f_c (for the HPF, to observe near-differentiator behaviour).
2. Observe and sketch $v_{in}(t)$ and $v_{out}(t)$ simultaneously on the CRO.
3. Compare the observed waveforms with Figures 3.4 and 3.5.

3.8 Tabulation

Low-pass filter.

Frequency (Hz)	V_{out} (V)	Gain = $20 \log_{10}(V_{out}/V_{in})$ (dB)	Phase (deg)
10			
100			
500			
1000			
2000			
5000			
10000			
50000			

High-pass filter.

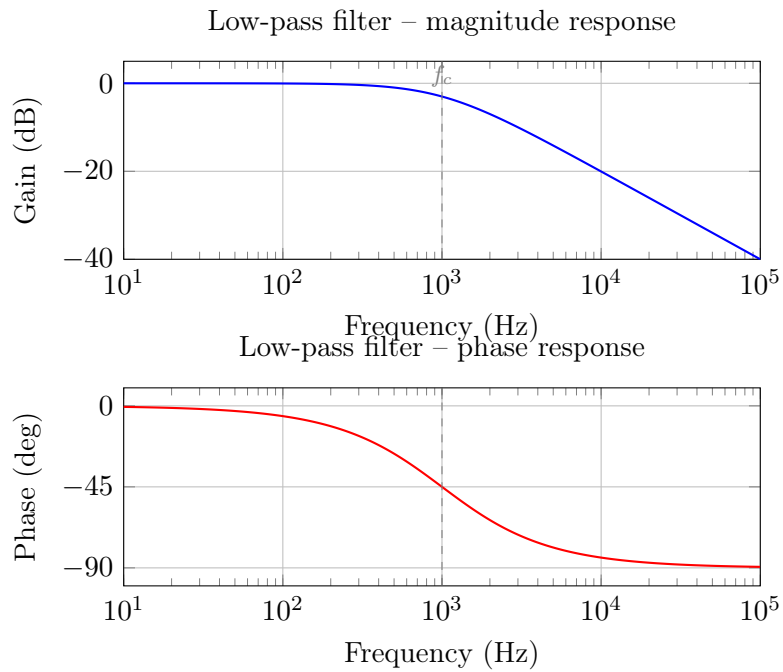
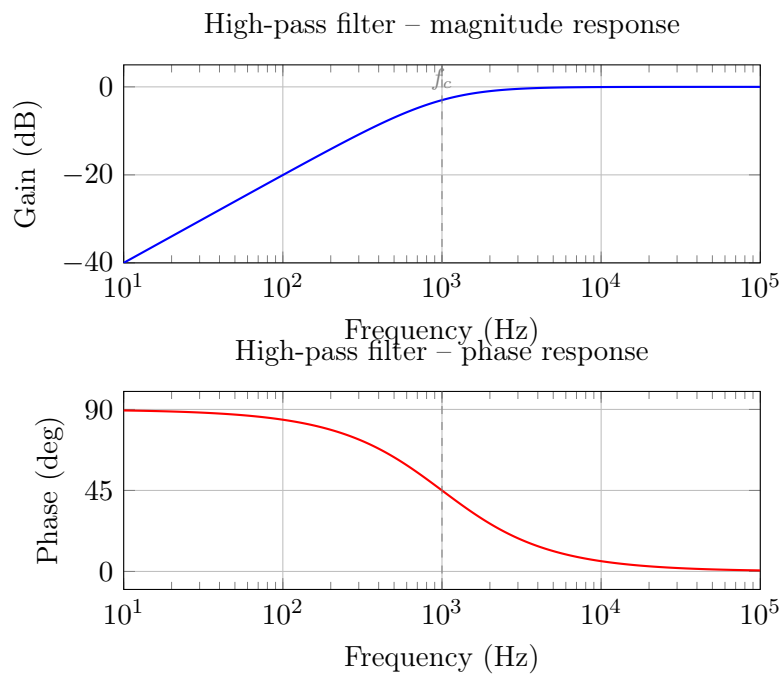
Frequency (Hz)	V_{out} (V)	Gain = $20 \log_{10}(V_{out}/V_{in})$ (dB)	Phase (deg)
10			
100			
500			
1000			
2000			
5000			
10000			
50000			

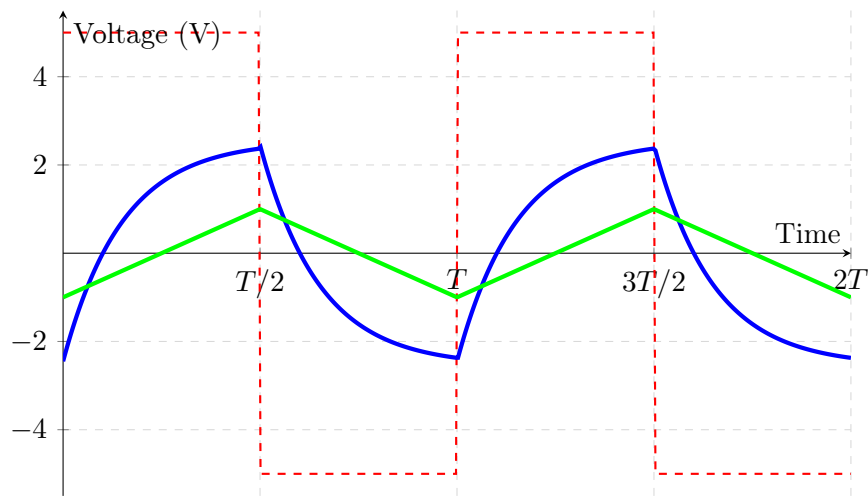
Bode Diagrams

The frequency-response curves (gain and phase Vs frequency) are known as Bode diagrams (after the inventor Hendrik Wade Bode). The y-axis of the gain plot uses the dB unit, and the phase plot uses the phase lag/lead in degrees. The x-axis is in log-scale, and frequencies in decades are marked. The sample Bode diagrams are as shown in Figs. 3.2 and 3.3.

3.9 Sample Waveforms (Square Wave Input)

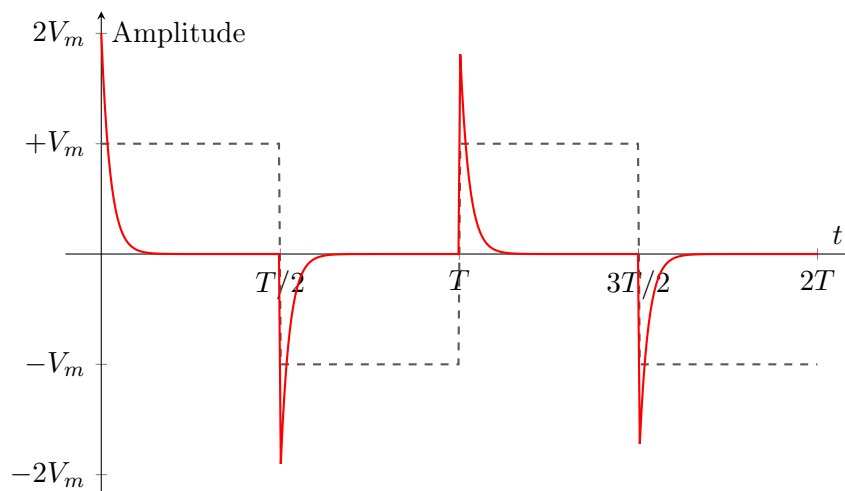
The time-domain behaviour of the RC circuit can be understood by studying the response to a square-wave of variable frequency, the waveforms of which are shown in Figs. 3.4 and 3.5. It can be seen that for $RC \gg T/2$, the RC low-pass filter circuit acts as an integrator, whereas for the RC high-pass circuit (with C and R interchanged), behaves as a differentiator when $RC \ll T/2$. Further, note that the waveform given in Fig. 3.5 are obtained under the constraint that $R \gg R_{out}$ of the source (function generator).

Figure 3.2: Bode plot of the RC low-pass filter ($f_c \approx 1$ kHz)Figure 3.3: Bode plot of the RC high-pass filter ($f_c \approx 1$ kHz)



--- Input V_{in} — Output V_{out} for $RC < T$ — Output V_{out} for $RC \gg T$

Figure 3.4: Low-pass filter response to a square wave (rounded/ramped output)



--- Input square wave — HPF output ($RC \ll T$: differentiator)

Figure 3.5: High-pass filter response to a square wave (decaying spikes at transitions)

3.10 Result

First-order RC low-pass and high-pass filters were designed for a cutoff frequency of 1 kHz using $R = 1.5 \text{ k}\Omega$ and $C = 0.1 \text{ }\mu\text{F}$. Their frequency response (Bode plots) and time-domain behaviour with a square-wave input were studied and verified experimentally against the theoretical response.

Explore further

The output impedance of the source (function generator) has an impact on practical measurements. Hence, while checking the output waveforms of the high-pass (or differentiator) circuit, the resistance value should be much higher than that of the output impedance of the function generator (typically, $50 \text{ }\Omega$). For satisfying the condition $RC \ll T/2$, we may choose a higher R and a lower C , or vice-versa. Try with both designs and see how the waveforms differ in each case practically, and explain why. What is the effect in the case of the integrator? When simulating, you may experiment with/without a source resistance to see the effects.

3.11 Questions

1. Derive the transfer function of a first-order RC low-pass filter.
2. Why do the LPF and HPF built from the same R and C have the same cutoff frequency?
3. What is the significance of the -3 dB point?
4. What is the asymptotic roll-off rate of a first-order filter, in dB/decade ?
5. Under what condition does an RC low-pass filter behave as an integrator, and a high-pass filter as a differentiator?
6. Why is the RC circuit (both LPF and HPF) in this experiment called “first-order” filters?
7. How would you extend this circuit to a second-order filter, and what advantage would that give?
8. What are the effects of function generator output impedance on the design/expected waveforms of the RC circuit?
9. How do the integrator and differentiator circuits behave when the input signal has a DC added to it ($\pm V_{dc}$)?

Experiment 4

RC-Coupled CE Amplifier using BJT

4.1 Aim

To design, assemble, and test a single-stage RC-coupled amplifier using a BJT in Common-Emitter (CE) configuration. The specific objectives are:

1. To verify the DC operating point (Q-point) of the transistor.
2. To measure the mid-band voltage gain.
3. To plot its frequency response curve (Gain in dB vs Frequency) on a semi-logarithmic scale.
4. To determine the lower cutoff frequency (f_L), upper cutoff frequency (f_H), and overall bandwidth (BW).

4.2 Pre-Lab Assignment

- (i) Given the supply voltage V_{CC} , desired quiescent collector current I_C , and BJT parameters (β , $V_{BE(on)}$) assigned to your batch, design the voltage-divider bias network and calculate R_1 , R_2 , R_C , R_E for a stable operating point.
- (ii) Calculate the expected mid-band voltage gain $A_{v(mid)}$ using the design values.
- (iii) Simulate the designed circuit in LTspice (Pre-Lab 2): run a `.ac` analysis and obtain the Bode plot (magnitude in dB and phase vs. frequency). Note the lower cutoff frequency f_L , upper cutoff frequency f_H , and bandwidth $BW = f_H - f_L$ from the simulated plot.
- (iv) Tabulate the theoretically designed values and the LTspice-simulated values for comparison with hardware measurements to be taken in this experiment.

4.3 Components and Equipment Required

Table 4.1 lists the components/equipment required.

4.4 Theory

A common-emitter (CE) configuration is widely used in voltage amplification due to its high voltage and current gain, along with moderate input and output impedances. In an RC-coupled amplifier, multiple single-stage amplifiers can be cascaded using a coupling capacitor, though this experiment focuses on analyzing a single-stage implementation. The capacitors C_{in} and C_{out} couple the AC signal into and out of the stage while blocking DC, isolating the amplifier's bias point from the source and the load; an emitter bypass capacitor C_E provides an AC ground at the emitter to maximise gain, while a DC bias network (R_1, R_2, R_E) fixes a stable quiescent operating point independent of transistor-to-transistor β variation.

S.No.	Component/Equipment	Specification
1	NPN Transistor	BC547 (or equivalent, e.g. BC107), 1 no.
2	Resistors	R_1, R_2, R_C, R_E as per design
3	Coupling Capacitors	C_{in}, C_{out} as per design
4	Bypass Capacitor	C_E as per design
5	Regulated DC Power Supply	0–15 V
6	Digital Multi-Meter	0–20 V
7	Function Generator	Sine wave, 1 Hz – 1 MHz
8	Digital Storage Oscilloscope	2-channel, 20–50 MHz bandwidth
9	Breadboard and connecting wires	–

Table 4.1: Components and equipment list for RC-coupled BJT amplifier

4.4.1 Circuit Functionality

- **Biasing Network (R_1, R_2, R_C, R_E):** Voltage-divider biasing (R_1 and R_2) is utilized to provide a stable operating point (Q -point) independent of temperature variations and transistor parameter (β) spreads. R_C serves as the collector load resistor across which the amplified AC output is developed. R_E provides negative feedback to stabilize the DC operating point.
- **Coupling Capacitors (C_{in}, C_{out}):** C_{in} blocks any DC component from the AC signal source, preventing it from altering the Q -point. Similarly, C_{out} couples the amplified AC signal to the load while blocking the DC voltage at the collector.
- **Bypass Capacitor (C_E):** C_E is placed in parallel with R_E to provide a low-impedance path for the AC signal to ground. This prevents AC degeneration across R_E , maximizing the mid-band voltage gain.

4.4.2 Frequency Response Characteristics

The performance of the amplifier over a wide frequency spectrum is characterized by three distinct bands:

1. **Low-Frequency Region:** At low frequencies, the capacitive reactances of C_{in}, C_{out} , and C_E (given by $X_C = 1/(2\pi fC)$) become significantly high. These reactances drop a considerable portion of the AC signal, thereby reducing the effective voltage gain.
2. **Mid-Band Region:** In this region (typically 100 Hz to 100 kHz), both the coupling capacitors behave as AC short circuits and the emitter bypass capacitor provides a low-impedance AC path, so the emitter is at AC ground. Concurrently, the internal parasitic capacitances of the BJT act as open circuits. Consequently, the voltage gain remains constant and reaches its maximum value ($A_{v(mid)}$). The small-signal mid-band gain, using the hybrid- π/r_e model, is

$$A_{v(mid)} = -g_m(R_C \parallel R_L) = -\frac{(R_C \parallel R_L)}{r_e}, \quad r_e \approx \frac{V_T}{I_C}, \quad (4.1)$$

where $V_T \approx 26$ mV at room temperature and R_L is the external load (e.g., the DSO probe or oscilloscope input impedance or the explicit load resistance added at the output). The negative sign indicates the characteristic 180° phase inversion of the CE stage.

3. **High-Frequency Region:** At high frequencies, the coupling and bypass capacitors continue to act as short circuits. However, the internal shunt parasitic capacitances of the

transistor (such as base-emitter capacitance C_{be} and base-collector capacitance C_{bc}) and stray wiring capacitances become prominent. Their reactances drop, shunting the signal to ground and causing the gain to fall.

The bandwidth (BW) is defined as the frequency range over which the voltage gain remains above 70.7% of its maximum mid-band value, corresponding to a 3 dB drop:

$$BW = f_H - f_L \quad (4.2)$$

where f_L is the lower cutoff frequency and f_H is the upper cutoff frequency.

4.5 Circuit Diagram

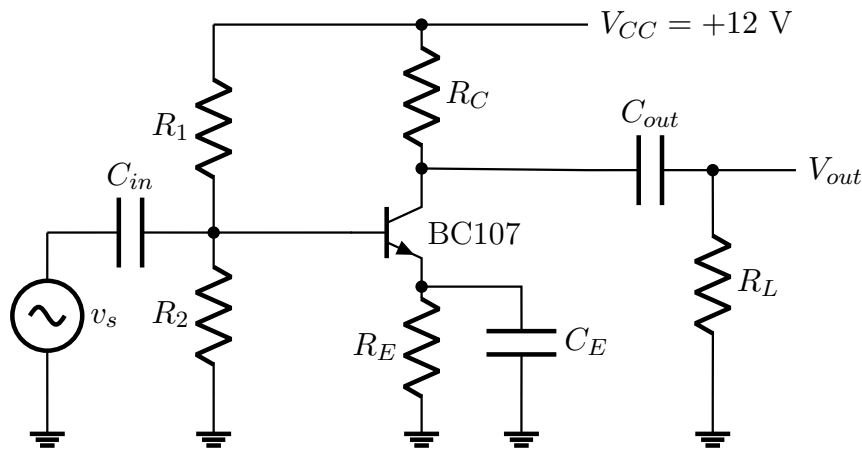


Figure 4.1: Single-stage RC-coupled CE amplifier with voltage-divider bias, emitter bypass capacitor C_E , and coupling capacitors C_{in} , C_{out} .

4.6 Design

4.6.1 Design Specifications

Let us design the amplifier for the following typical specifications:

- Supply Voltage, $V_{CC} = 12\text{ V}$
- Collector Current, $I_C = 2\text{ mA}$
- Transistor Current Gain, $h_{fe} = \beta = 100$
- Transistor Base-Emitter Voltage, $V_{BE} = 0.7\text{ V}$

4.6.2 Step-by-Step Design Procedure

1. **Design of R_E :** For good thermal stability, allocate approximately 10% of V_{CC} across the emitter resistor R_E :

$$V_E = 0.1 \times V_{CC} = 0.1 \times 12\text{ V} = 1.2\text{ V} \quad (4.3)$$

Since $I_E \approx I_C = 2\text{ mA}$:

$$R_E = \frac{V_E}{I_E} = \frac{1.2\text{ V}}{2\text{ mA}} = 600\ \Omega \quad (\text{Select standard value: } 560\ \Omega) \quad (4.4)$$

2. **Design of R_C :** For maximum symmetrical output swing, select the quiescent collector-emitter voltage $V_{CE} = 0.5 \times V_{CC} = 6 \text{ V}$. Applying Kirchhoff's Voltage Law (KVL) to the collector loop:

$$V_{CC} = I_C R_C + V_{CE} + V_E \quad (4.5)$$

$$R_C = \frac{V_{CC} - V_{CE} - V_E}{I_C} = \frac{12 - 6 - 1.2}{2 \text{ mA}} = \frac{4.8 \text{ V}}{2 \text{ mA}} = 2.4 \text{ k}\Omega \quad (\text{Select standard value: } 2.2 \text{ k}\Omega) \quad (4.6)$$

3. **Design of Biasing Network (R_1 and R_2):** The base voltage V_B is given by:

$$V_B = V_E + V_{BE} = 1.2 \text{ V} + 0.7 \text{ V} = 1.9 \text{ V} \quad (4.7)$$

The base current is $I_B = I_C / \beta = 2 \text{ mA} / 100 = 20 \mu\text{A}$. For stable voltage divider operation, the current flowing through R_2 (I_2) should be much larger than I_B . Typically, $I_2 \geq 10 \times I_B$:

$$I_2 = 10 \times 20 \mu\text{A} = 200 \mu\text{A} \quad (4.8)$$

Therefore:

$$R_2 = \frac{V_B}{I_2} = \frac{1.9 \text{ V}}{200 \mu\text{A}} = 9.5 \text{ k}\Omega \quad (\text{Select standard value: } 10 \text{ k}\Omega) \quad (4.9)$$

The current through R_1 is $I_1 = I_2 + I_B = 200 \mu\text{A} + 20 \mu\text{A} = 220 \mu\text{A}$:

$$R_1 = \frac{V_{CC} - V_B}{I_1} = \frac{12 - 1.9}{220 \mu\text{A}} = \frac{10.1 \text{ V}}{220 \mu\text{A}} \approx 45.91 \text{ k}\Omega \quad (\text{Select standard value: } 47 \text{ k}\Omega) \quad (4.10)$$

4. **Selection of load resistor, R_L :** From Eqn. 4.1, $r_e \approx \frac{26 \text{ mV}}{2 \text{ mA}} = 12 \Omega$.

For a gain of 50, substituting in Eqn. 4.1, we get, $R_L = 825 \Omega$.

Select standard value of 820Ω for R_L .

5. **Design of Capacitors (C_{in} , C_{out} , C_E):** To ensure low frequency signals are not attenuated, capacitive reactances are specified at the lower cutoff boundary (assume $f_L = 100 \text{ Hz}$):

- The emitter bypass capacitor C_E must satisfy $X_{CE} \leq R_E / 10$:

$$C_E \geq \frac{1}{2\pi \cdot f_L \cdot (R_E / 10)} = \frac{1}{2\pi \cdot 100 \cdot 56} \approx 28.4 \mu\text{F} \quad (\text{Select standard value: } 33 \mu\text{F}) \quad (4.11)$$

- Input and output coupling capacitors (C_{in} , C_{out}) are typically chosen to ensure proper signal transmission into the base impedance and load. Select C_{in} , and C_{out} , so that their reactances are negligible ($X_C \leq R / 10$, where R is the resistance they work against - $R = R_1 \parallel R_2 \parallel h_{fe} r_e$ for C_{in} , and $R = R_L$ for C_{out}) at the desired lower cutoff frequency f_L :

$$C \geq \frac{1}{2\pi f_L (R/10)} \quad (4.12)$$

Then, for C_{in} :

$$R = R_1 \parallel R_2 \parallel h_{fe} r_e = 1047.5 \Omega$$

$$C_{in} \geq \frac{1}{2\pi \cdot 100 \cdot 104.75} = 15.19 \mu\text{F}$$

Use standard capacitor values rounded up from the calculated minimum. Standard value of $22 \mu\text{F}$ may be selected.

For C_{out} :

$$R = R_L = 820 \Omega$$

$$C_{out} \geq \frac{1}{2\pi \cdot 100 \cdot 82} = 19.4 \mu\text{F}$$

Standard value of $22 \mu\text{F}$ may be selected.

- Verify (via LTspice, as in the Pre-Lab) that the resulting f_L , f_H , and $A_{v(mid)}$ meet the target specification; iterate the design if necessary.

4.7 Procedure

The circuit is set up as shown in Figure 4.1. First, check the DC bias voltages at various nodes (Base, Collector, Emitter etc., with respect to ground. Record the observations in Table 4.2. For frequency response, an input sinusoidal signal (V_{in}) from a function generator with amplitude 20 mV_{p-p} is given to the circuit. Measure the magnitude (peak to peak) of the input by using DSO CH1. Connect the DSO CH2 to the output side and the amplified output is observed. Increase the frequency in steps and observe the magnitude of V_{out} . The frequency response is plotted in a semi log sheet.

4.8 Precautions

- **Input Signal Amplitude Control:** The input voltage must be kept very low (tens of millivolts). Introducing a large input signal will drive the transistor into saturation and cutoff, causing severe clipping distortion in the output waveform and invalidating gain calculations.
- Ensure correct transistor pin identification (Emitter, Base, Collector) before insertion; an incorrectly connected transistor may get damaged.
- Switch off the DC supply before making or changing any circuit connections.
- While sweeping frequency, allow the output to stabilise before recording each reading, especially near the cutoff frequencies.
- **Common Grounding:** Ensure that the grounds of the power supply, function generator, and oscilloscope channels are securely connected to a single common ground rail on the breadboard to eliminate noise and ground-loop interference.
- **Capacitor Polarities:** Pay close attention to the polarity marks of the electrolytic capacitors (C_{in} , C_{out} , C_E). The positive terminals must always face nodes with higher DC potential (C_{in} positive terminal to BJT Base; C_{out} positive terminal to BJT Collector; C_E positive terminal to BJT Emitter).

4.9 Model Graph / Expected Waveforms

The expected frequency response curve for the designed RC-coupled CE amplifier is shown in Figure 4.2

4.10 Observations

4.10.1 DC Operating Point Parameters

Record the measured static DC parameters in Table 4.2 and compare them with the calculated theoretical values.

4.10.2 AC Frequency Response Data

Record the input and output peak-to-peak voltages across the frequency range in Table 4.3. Maintain V_{in} constant.

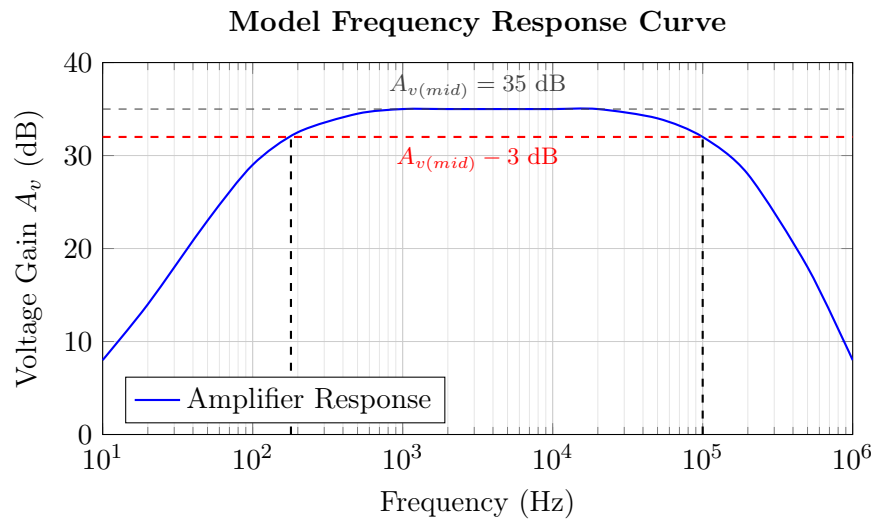


Figure 4.2: Semi-logarithmic plot of the Expected Frequency Response

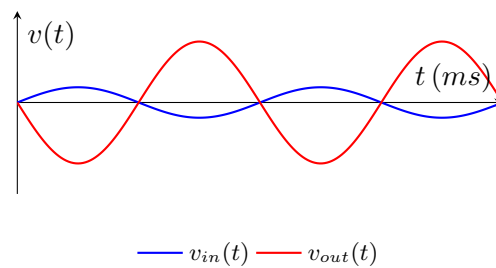


Figure 4.3: Expected input and (amplified, inverted) output waveforms in the mid-band region.

Parameter	Theoretical Value	Measured Value
Base Voltage (V_B)		
Emitter Voltage (V_E)		
Collector Voltage (V_C)		
Collector-Emitter Voltage (V_{CEQ})		
Collector Current (I_{CQ})		

Table 4.2: DC Operating Point Measurements

S. No.	Freq. f (Hz)	$V_{in,pp}$ (V)	$V_{out,pp}$ (V)	$A_v = V_{out,pp}/V_{in,pp}$	$A_v = 20 \log_{10}(A_v)$ (dB)
1	10				
2	20				
⋮	⋮				
⋮	⋮				
⋮	⋮				
	500k				
	1M				

Table 4.3: Frequency response data (add rows as needed to adequately capture the roll-off regions).

4.11 Result

The RC-coupled CE amplifier was designed, assembled, and tested. The performance parameters of the single-stage RC-coupled BJT amplifier were experimentally measured and determined from the frequency response plot:

- Measured Mid-Band Voltage Gain, $A_{v(mid)} = \underline{\hspace{2cm}}$ (or $\underline{\hspace{2cm}}$ dB)
- Lower Cutoff Frequency, $f_L = \underline{\hspace{2cm}}$ Hz
- Upper Cutoff Frequency, $f_H = \underline{\hspace{2cm}}$ Hz
- Experimental Bandwidth ($BW = f_H - f_L$) = $\underline{\hspace{2cm}}$ Hz

The DC operating point was successfully verified, and the plotted frequency response reflects the expected drop in voltage gain at low and high frequencies.

4.12 Questions

- Q1. Why is voltage-divider bias preferred over fixed bias for BJT amplifier stages in terms of Q-point stability against β variation?
- Q2. Explain, with reasoning, why the emitter bypass capacitor C_E is essential for achieving high mid-band gain, and what happens to $A_{v(mid)}$ if C_E is removed.
- Q3. Derive the expression for the lower cutoff frequency contributed by C_{in} alone, treating all other capacitors as ideal AC shorts.
- Q4. What is the Miller effect, and how does it contribute to the reduction in gain at high frequencies in a CE amplifier?
- Q5. Why must the input signal amplitude be kept small during this experiment? What form of distortion would be observed if it were too large, and why?
- Q6. Compare the phase relationship between v_{in} and v_{out} in the CE amplifier with that of the RC network studied in Pre-Lab 1.
- Q7. What would happen to the output signal if the input coupling capacitor C_{in} developed an internal short circuit?
- Q8. How does the load resistance R_L affect the mid-band voltage gain of the amplifier?
- Q9. Why is the common-emitter configuration the preferred choice for cascading multiple amplification stages compared to common-base or common-collector configurations?
- Q10. What is the function of the input coupling capacitor?
- Q11. Why is the emitter bypass capacitor used?
- Q12. Why does the output waveform undergo phase reversal in CE configuration?
- Q13. What is meant by midband gain?
- Q14. Define lower and upper cut-off frequencies.
- Q15. Why does gain decrease at low frequencies?
- Q16. Why does gain decrease at high frequencies?

- Q17. What is the significance of the -3 dB point?
- Q18. How is bandwidth related to the frequency response curve?
- Q19. Why should the transistor be properly biased before applying the input signal?
- Q20. What type of coupling is used between stages in an RC-coupled amplifier?
- Q21. How does changing R_C affect the gain of the amplifier?
- Q22. Name one limitation of an RC-coupled amplifier.

Experiment 5

MOSFET Amplifier

5.1 Aim

To design, set up, and test a single-stage Common Source (CS) MOSFET amplifier using a voltage-divider bias configuration. The specific objectives are:

1. To determine and verify the DC operating point (Q -point) of the MOSFET.
2. To measure the mid-band voltage gain of the CS amplifier.
3. To plot the frequency response curve (Gain in dB vs Frequency) on a semi-logarithmic scale.
4. To evaluate the lower cutoff frequency (f_L), upper cutoff frequency (f_H), and overall bandwidth (BW).

5.2 Pre-Lab Assignment

Before performing the experiment, students must complete the following analytical tasks:

1. For the circuit specified in the Design section ($V_{DD} = 12\text{ V}$, $R_1 = 1\text{ M}\Omega$, $R_2 = 470\text{ k}\Omega$, $R_D = 2.7\text{ k}\Omega$, $R_S = 560\text{ }\Omega$), calculate the theoretical DC operating point parameters (V_G , V_S , I_D , and V_{DS}). Assume the N-channel MOSFET has a threshold voltage $V_{GS(th)} = 2.0\text{ V}$ and conduction parameter $K_n = 2.0\text{ mA V}^{-2}$.
2. Sketch the small-signal equivalent AC circuit of the Common Source amplifier and derive the expression for the mid-band voltage gain (A_v).
3. Compare a MOSFET Common Source amplifier with a BJT Common Emitter amplifier in terms of input impedance, output impedance, and voltage gain.
4. Explain why the source bypass capacitor (C_S) is necessary and how its removal would impact the amplifier's input impedance and voltage gain.

5.3 Components and Equipment Required

The components and test instruments required for this experiment are listed in Table [5.1](#).

5.4 Theory

The Common Source (CS) configuration of a Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is highly valued in analog electronics primarily due to its exceptionally high input impedance, which approaches infinity at DC and low frequencies. This characteristic prevents the amplifier stage from loading down the preceding signal source.

Table 5.1: Components and Equipment Specifications

S. No.	Item Name	Specification / Range	Quantity
1	MOSFET	N-Channel Enhancement (2N7000 or BS170)	1 No.
2	Resistors	1 M Ω , 470 k Ω , 2.2 k Ω , 560 Ω , 10 k Ω (Load)	1 No. each
3	Capacitors	1 μ F (Electrolytic), 47 μ F (Electrolytic)	2 Nos., 1 No.
4	Dual Regulated Power Supply	0 – 30 V, 1 A (DC)	1 No.
5	Function Generator	1 Hz – 1 MHz, Sinusoidal output	1 No.
6	Digital Storage Oscilloscope (DSO)	Dual channel, 20 MHz or higher	1 No.
7	Breadboard & Connecting Wires	Standard multi-contact, single-strand wires	As required

5.4.1 Circuit Functionality

- **Voltage-Divider Biasing (R_1, R_2):** Since the insulated gate of a MOSFET draws virtually zero DC current ($I_G \approx 0$), the gate voltage V_G is established purely by the voltage divider network across the supply rail V_{DD} . This arrangement provides an exceptionally stable operating point.
- **Drain and Source Resistors (R_D, R_S):** R_D dictates the quiescent drain voltage and converts the dynamic AC drain current into an amplified voltage signal. R_S provides self-bias stabilization against device parameter variations.
- **Coupling and Bypass Capacitors (C_{in}, C_{out}, C_S):** C_{in} and C_{out} isolate the DC bias voltages of the stage from the input source and output load. The source bypass capacitor C_S provides an AC short circuit across R_S to ground, preventing negative feedback (degeneration) from reducing the mid-band AC voltage gain.

5.4.2 Frequency Response Characteristics

Similar to other RC-coupled architectures, the voltage gain drops off at extreme frequencies:

1. **Low-Frequency Roll-off:** Dominated by the increasing series reactances of C_{in} and C_{out} , along with the parallel impedance of C_S . As the frequency approaches DC, these components attenuate the signal or introduce internal degeneration.
2. **Mid-Band Plateau:** The coupling/bypass capacitors act as optimal short circuits while internal parasitic capacitances remain open. The voltage gain remains flat at its maximum value, given by:

$$A_{v(mid)} \approx -g_m(R_D \parallel R_L) \quad (5.1)$$

where g_m represents the transconductance of the MOSFET at its given operating point.

3. **High-Frequency Roll-off:** Caused by internal device capacitances (gate-to-source C_{gs} , gate-to-drain C_{gd}) and stray wiring capacitance. The effective input capacitance is heavily multiplied due to the Miller effect, shunting high-frequency signals to ground.

The system bandwidth is evaluated using the upper (f_H) and lower (f_L) frequencies where the gain falls by 3 dB (70.7%) from its mid-band value:

$$BW = f_H - f_L \quad (5.2)$$

5.5 Circuit Diagram

The schematic for the single-stage voltage-divider biased CS MOSFET amplifier is presented in Figure 5.1.

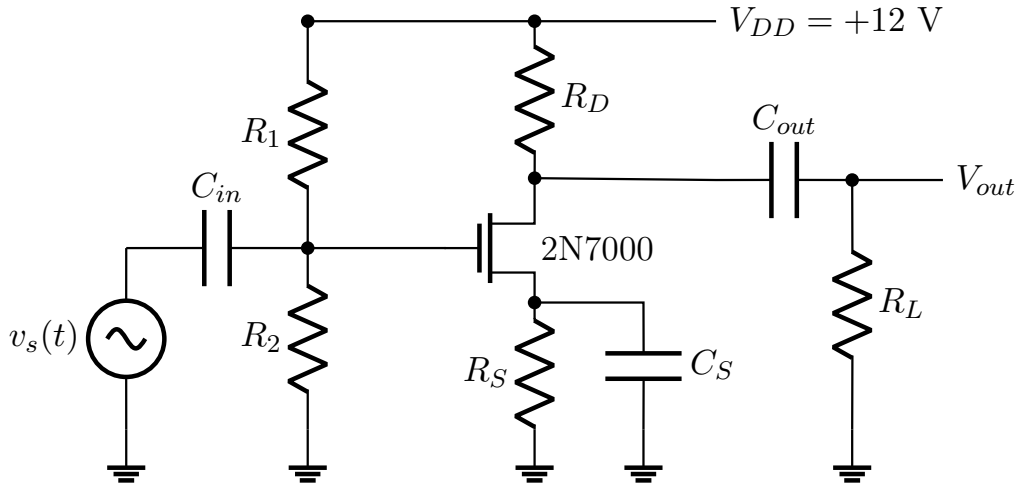


Figure 5.1: Circuit Diagram of a Single-Stage Voltage-Divider Biased CS MOSFET Amplifier

5.6 Design

5.6.1 Design Specifications

Let us design the amplifier for the following typical experimental parameters:

- Supply Voltage, $V_{DD} = 12\text{ V}$
- Quiescent Drain Current, $I_D = 2\text{ mA}$
- MOSFET Conduction Parameter, $K_n = 2\text{ mA V}^{-2}$
- MOSFET Threshold Voltage, $V_{GS(th)} = 2.0\text{ V}$

5.6.2 Step-by-Step Design Procedure

1. **Design of R_S :** To safeguard against thermal runaway and track variations smoothly, allocate roughly 10% of V_{DD} across the source resistor R_S :

$$V_S = 0.1 \times V_{DD} = 0.1 \times 12\text{ V} = 1.2\text{ V} \quad (5.3)$$

Given $I_D = 2\text{ mA}$:

$$R_S = \frac{V_S}{I_D} = \frac{1.2\text{ V}}{2\text{ mA}} = 600\ \Omega \quad (\text{Select standard value: } 560\ \Omega) \quad (5.4)$$

2. **Design of R_D :** For balanced signal swing, aim to drop approximately half of the remaining supply headroom across the drain resistor R_D :

$$V_{RD} = \frac{V_{DD} - V_S}{2} = \frac{12\text{ V} - 1.2\text{ V}}{2} = 5.4\text{ V} \quad (5.5)$$

$$R_D = \frac{V_{RD}}{I_D} = \frac{5.4\text{ V}}{2\text{ mA}} = 2.7\text{ k}\Omega \quad (\text{Select standard value: } 2.7\text{ k}\Omega). \quad (5.6)$$

3. Design of Biasing Network (R_1 and R_2):

The transfer characteristics of 2N7000 gives:

$$V_{GS} \approx 2.3 - 2.7 \text{ V}$$

for $I_D = 2 \text{ mA}$.

Take $V_{GS} \approx 2.5 \text{ V}$.

Required gate voltage

$$V_G = V_S + V_{GS} = 1.2 \text{ V} + 2.5 \text{ V} = 3.5 \text{ V}$$

Choose divider current of approximately $20 \mu\text{A}$.¹

Then,

$$R_1 + R_2 = \frac{12 \text{ V}}{20 \mu\text{A}} \approx 600 \text{ k}\Omega$$

$$\frac{R_2}{R_1 + R_2} = \frac{3.5}{12} = 0.292$$

$R_2 = 0.292 \times 600 \text{ k} = 175 \text{ k}$; Choose a standard value of $R_2 = 180 \text{ k}\Omega$.

Then $R_1 = 600 \text{ k} - 180 \text{ k} = 420 \text{ k}$. Choose a standard value of $R_1 = 470 \text{ k}\Omega$.

5.7 Procedure

Set up the circuit as shown in the Figure 5.1. Before giving the input ac signal, check the bias voltages. Apply an input sinusoidal signal of 40 mV (peak-to-peak) at 1000 Hz. Observe the output on the DSO. Vary the frequency of the input signal over a range of values (from 50Hz to a few MHz) to obtain the frequency response which is a graph between $\log f$ (x-axis) and gain in dB (y-axis) (See Table 5.3).

5.8 Precautions

1. **Electrostatic Discharge (ESD) Awareness:** MOSFETs possess extremely thin insulated gate oxide barriers that are highly vulnerable to static electricity. Ground your hands before handling the transistor pins and do not insert or remove components while the power supply is turned on.
2. **Signal Compression Constraints:** Ensure V_{in} does not drive the output near the saturation or cutoff boundaries. If the output peaks begin to flatten, immediately scale down the signal generator's output level.
3. **Bypass Capacitor Integrity:** Double-check that the electrolytic capacitors are oriented correctly with respect to node potentials to avoid leakage currents destabilizing the operating point.

5.9 Model Graph / Expected Waveforms

The typical performance characteristic curve for the configured CS amplifier is presented in Figure 5.2.

¹For higher input impedance, a lower divider current may be assumed, which results in higher divider resistance values.

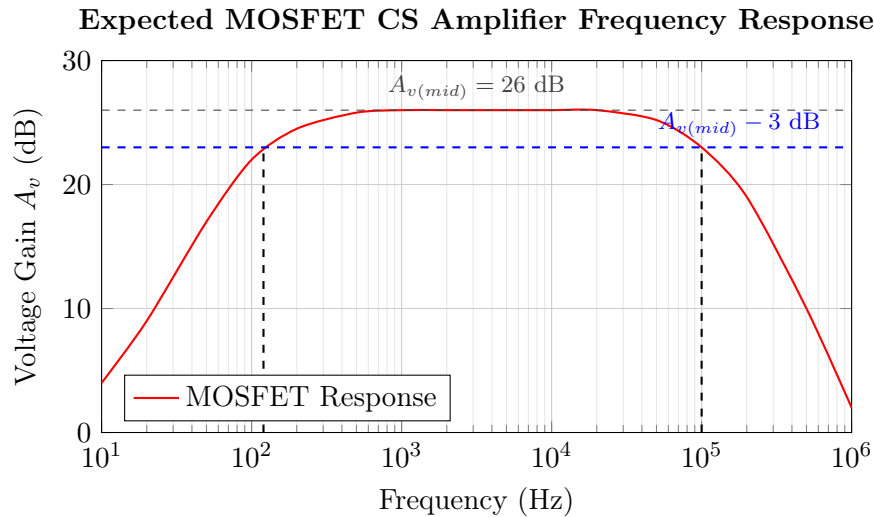


Figure 5.2: Semi-logarithmic plot of the Expected Frequency Response

5.10 Observation Tables

5.10.1 DC Operating Point Parameters

Compare the static baseline target parameters against actual laboratory multimeter readings in Table 5.2.

Table 5.2: DC Biasing Values

Parameter	Theoretical Value	Practical Value (Measured)
Gate Voltage (V_G)		
Source Voltage (V_S)		
Drain Voltage (V_D)		
Gate-Source Voltage (V_{GS})		
Drain-Source Voltage (V_{DS})		
Drain Current (I_D)		

5.10.2 AC Frequency Response Data

Log the raw output amplitudes across the sweep spectrum in Table 5.3, ensuring V_{in} remains fixed.

5.11 Result

The specific performance metrics of the voltage-divider biased single-stage Common Source MOSFET amplifier were successfully mapped and derived:

- Maximum Mid-Band Voltage Gain, $A_{v(mid)} = \underline{\hspace{2cm}}$ (or $\underline{\hspace{2cm}}$ dB)
- Lower Cutoff Frequency, $f_L = \underline{\hspace{2cm}}$ Hz
- Upper Cutoff Frequency, $f_H = \underline{\hspace{2cm}}$ Hz
- Measured Amplifier Bandwidth ($BW = f_H - f_L$) = $\underline{\hspace{2cm}}$ Hz

S. No.	Freq. f (Hz)	$V_{in,pp}$ (V)	$V_{out,pp}$ (V)	$A_v = V_{out,pp}/V_{in,pp}$	$A_v = 20 \log_{10}(A_v)$ (dB)
1	10				
2	20				
$\vdots$	$\vdots$				
$\vdots$	$\vdots$				
$\vdots$	$\vdots$				
	500k				
	1M				

Table 5.3: AC Frequency Sweep Measurements

The baseline Q -point parameters were verified to support operation within the saturation region, and the frequency response confirms high-frequency roll-off due to parasitic device effects.

5.12 Questions

1. Why does the input impedance of a MOSFET amplifier drop when operating at very high frequencies?
2. What is meant by the term "Transconductance" (g_m) of a MOSFET, and how does it affect the voltage gain?
3. Explain the criteria required to ensure an enhancement-mode N-channel MOSFET remains in its saturation region for large signal swings.
4. How does adding a load resistor (R_L) in parallel with the output alter the upper and lower cutoff frequencies?
5. What is the significance of the Miller effect, and which parasitic internal capacitance is responsible for dragging down the high-frequency response in a CS amplifier?

Experiment 6

Inverting and Non-Inverting Amplifiers Using Op-amps

6.1 Aim

To design, set up, and analyze Operational Amplifier (Op-Amp) based Inverting and Non-Inverting Amplifiers using IC 741. The specific objectives are:

1. To verify the closed-loop voltage gain for both inverting and non-inverting configurations.
2. To observe and record the input-output phase relationship for each configuration.

6.2 Pre-Lab Assignment

- (i) Review the concept of the *ideal op-amp* (infinite open-loop gain, infinite input impedance, zero output impedance) and the two golden rules of negative feedback analysis: (a) no current flows into the input terminals, and (b) the differential input voltage is zero (virtual short).
- (ii) For a given gain assigned to your batch, design suitable resistor values for both the inverting and non-inverting configurations.
- (iii) Simulate both designed circuits in LTspice (Pre-Lab 2) using the `OPAMP` or `UniversalOpAmp2` macro-model. Run a `.tran` analysis with a 1 kHz sine input and verify the expected gain and phase (inverted/non-inverted) at the output.

6.3 Components and Equipment Required

The required equipment and electronic components are listed in Table 6.1.

S.No.	Component/Equipment	Specification
1	Operational Amplifier IC	μ A741 (or equivalent), 1 no.
2	Resistors	1 k Ω , 10 k Ω , 22 k Ω
3	Dual Regulated DC Power Supply	± 15 V
4	Function Generator	Sine wave, 1 kHz
5	Digital Storage Oscilloscope	2-channel, 20–50 MHz bandwidth
6	Breadboard and connecting wires	–

Table 6.1: Components and equipment list.

6.4 Theory

The operational amplifier (op-amp) is a high-gain DC-coupled voltage amplifier with differential inputs and single-ended output. When combined with an external negative feedback network, it realises a wide range of precise linear functions, the most fundamental being the inverting and non-inverting amplifiers. Analysis of both configurations proceeds from the *ideal op-amp* assumptions:

- (a) The input bias currents are zero (infinite input impedance), so no current flows into either input terminal.
- (b) The open-loop gain A_{OL} is infinite, so with negative feedback the differential input voltage $v_+ - v_- \rightarrow 0$ (the *virtual short* condition).

6.4.1 Inverting Amplifier Configuration

In the inverting amplifier, the input signal v_{in} is applied to the inverting terminal (pin 2) through resistor R_1 , while the non-inverting terminal (pin 3) is connected to ground through a bias-compensating resistor R_{comp} . Negative feedback is established via feedback resistor R_f .

Due to the extremely high open-loop gain and high input impedance of the op-amp, the differential voltage across the input terminals is virtually zero ($v_d \approx 0$). Consequently, the inverting input is held at a potential equal to the non-inverting terminal (0 V), creating a **Virtual Ground**.

$$i_1 = \frac{v_{in} - 0}{R_1}, \quad i_f = \frac{0 - v_{out}}{R_f} \quad (6.1)$$

Since no current enters the ideal op-amp terminal ($i_1 = i_f$):

$$A_v = \frac{v_{out}}{v_{in}} = -\frac{R_f}{R_1} \quad (6.2)$$

The negative sign indicates a 180° phase shift between the input and output signals.

6.4.2 Non-Inverting Amplifier Configuration

In the non-inverting amplifier, the input signal v_{in} is applied directly to the non-inverting terminal (pin 3). The feedback network (R_1 and R_f) forms a voltage divider that feeds a portion of the output voltage back to the inverting terminal (pin 2).

Applying the virtual short concept ($v_- = v_+ = v_{in}$):

$$v_{in} = v_{out} \left(\frac{R_1}{R_1 + R_f} \right) \quad (6.3)$$

Rearranging gives the closed-loop voltage gain:

$$A_v = \frac{v_{out}}{v_{in}} = 1 + \frac{R_f}{R_1} \quad (6.4)$$

The positive sign confirms that the output waveform is in phase (0° phase shift) with the input signal.

6.5 Circuit Diagram

The circuit schematics for both configurations are detailed in Figures 6.1 and 6.2.

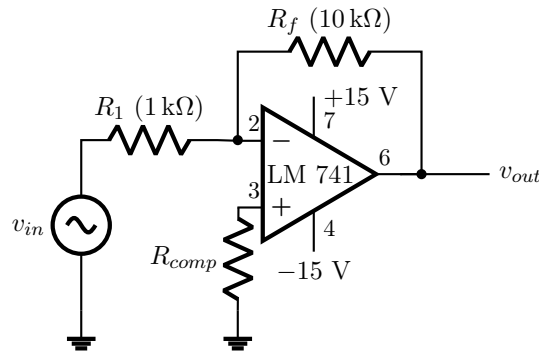


Figure 6.1: Inverting Amplifier Circuit Diagram

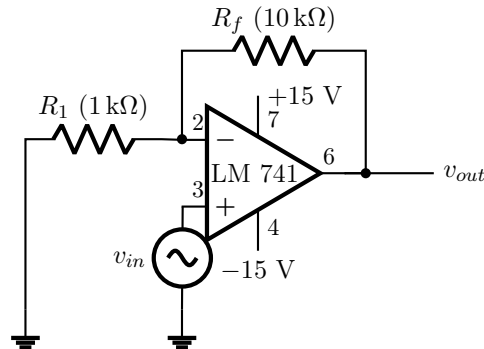


Figure 6.2: Non-Inverting Amplifier Circuit Diagram

6.6 Design

6.6.1 Inverting Amplifier Design ($|A_v| = 10$)

1. Choose an input resistor $R_1 = 1 \text{ k}\Omega$ to ensure negligible loading on the input signal source while keeping currents within normal limits.
2. Calculate the feedback resistor R_f :

We know that the gain

$$A_v = -\frac{R_f}{R_1}$$

$$|A_v| = \frac{R_f}{R_1} \implies 10 = \frac{R_f}{1 \text{ k}\Omega} \implies R_f = 10 \text{ k}\Omega \quad (6.5)$$

3. Calculate the offset-minimizing resistor R_{comp} connected to the non-inverting terminal to balance input bias currents:

$$R_{comp} = R_1 \parallel R_f = \frac{1 \text{ k}\Omega \cdot 10 \text{ k}\Omega}{1 \text{ k}\Omega + 10 \text{ k}\Omega} \approx 909 \Omega \quad (\text{Select standard value: } 1 \text{ k}\Omega) \quad (6.6)$$

6.6.2 Non-Inverting Amplifier Design ($A_v = 11$)

1. Select $R_1 = 1 \text{ k}\Omega$.
2. Determine R_f :

$$A_v = 1 + \frac{R_f}{R_1} \implies 11 = 1 + \frac{R_f}{1 \text{ k}\Omega} \implies R_f = 10 \text{ k}\Omega \quad (6.7)$$

6.7 Procedure

6.7.1 Inverting Amplifier Setup and Measurement

1. Connect +15 V DC to pin 7 ($+V_{CC}$) and -15 V DC to pin 4 ($-V_{EE}$) of IC 741 on the breadboard.
2. Construct the inverting amplifier circuit shown in Figure 6.1 with $R_1 = 1\text{ k}\Omega$ and $R_f = 10\text{ k}\Omega$.
3. Connect Channel 1 of the DSO across the input v_{in} and Channel 2 across the output v_{out} .
4. Set the function generator to produce a sine wave of 1.0 V_{p-p} at a frequency of 1 kHz.
5. Measure the peak-to-peak output voltage V_{out} and observe the phase relation between v_{in} and v_{out} .

6.7.2 Non-Inverting Amplifier Setup and Measurement

1. Reconfigure the circuit to form the non-inverting amplifier as shown in Figure 6.2.
2. Apply a 1.0 V_{p-p} sine wave at 1 kHz to the non-inverting input terminal (pin 3).
3. Measure V_{out} and observe the phase shift relative to v_{in} .

6.8 Precautions

1. Do not exceed the maximum supply voltage limits ($\pm 18\text{ V}$) for IC 741.
2. Reversed supply connections can permanently damage the IC.
3. Before applying input signal to the IC, power the IC.
4. Always establish a common ground connection between the power supply, signal generator, and oscilloscope.
5. To avoid saturation distortion, ensure that $V_{in} \cdot A_v$ does not exceed the op-amp saturation limits ($\approx \pm 13\text{ V}$).
6. Switch off the power supply before changing any circuit connections.

6.9 Model Graph / Expected Waveforms

Figure 6.3 and 6.4 shows the expected input-output waveforms illustrating the phase shifts for both configurations.

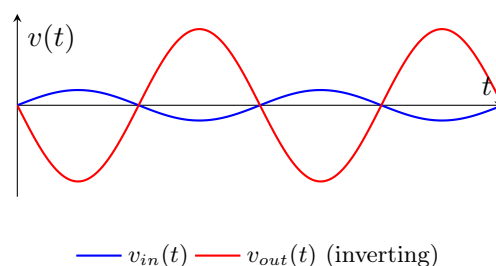


Figure 6.3: Expected waveforms for the inverting amplifier: v_{out} is amplified and 180° out of phase with v_{in} .

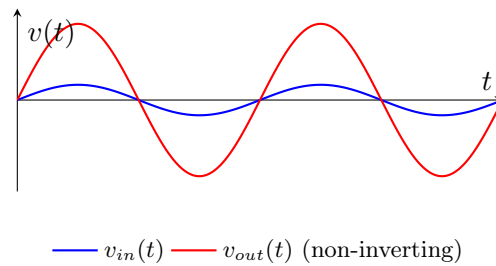


Figure 6.4: Expected waveforms for the non-inverting amplifier: v_{out} is amplified and in phase with v_{in} .

6.10 Result

The performance metrics of the Op-Amp Inverting and Non-Inverting Amplifiers were experimentally measured and analyzed:

6.11 Questions

1. What is meant by a "Virtual Ground" in op-amp circuits, and why does it not exist at the inverting pin of a non-inverting amplifier?
2. Why does a non-inverting amplifier exhibit a significantly higher input impedance than an inverting amplifier?
3. What happens to the output waveform if the input amplitude is increased such that $V_{in} \cdot A_v > V_{sat}$?
4. Explain how the Gain-Bandwidth Product (GBWP) limits the maximum operational frequency of an op-amp circuit when closed-loop gain is increased.
5. What is the function of the offset-nulling pins (pins 1 and 5) on the IC 741 operational amplifier?

Experiment 7

Adder and Subtractor Using Op-amps

7.1 Aim

To design, assemble, and experimentally verify the performance of Operational Amplifier (Op-Amp) mathematical circuits using IC 741. The specific objectives are:

1. To implement an Inverting Summing Amplifier (Adder) and verify that $V_{out} = -(V_1 + V_2)$.
2. To implement a Difference Amplifier (Subtractor) and verify that $V_{out} = V_2 - V_1$.
3. To observe and record the input and output waveforms for both DC and AC input signals.

7.2 Pre-Lab Assignment

1. Review the virtual-short and zero-input-current assumptions of the ideal op-amp and their application to circuits with multiple input signals.
2. For a summing amplifier with two/three inputs and a target output relationship assigned to your batch (e.g., $v_{out} = -(v_1 + v_2)$), design the required resistor values.
3. For a difference amplifier with a target gain assigned to your batch, design the required resistor values, ensuring the condition for common-mode rejection (Equation (??)) is satisfied.
4. Simulate both designed circuits in LTspice with appropriate DC and/or sinusoidal sources at each input, and verify the expected output algebraically combines the inputs as designed.

7.3 Components and Equipment Required

The components and test instruments required for this experiment are listed in Table 7.1.

S.No.	Component/Equipment	Specification
1	Operational Amplifier IC	$\mu A741$ (or equivalent), 1 no.
2	Resistors	R_1, R_2, R_3, R_f as per design
3	Dual Regulated DC Power Supply	± 15 V
4	Function Generator(s)	Sine wave, 1 Hz – 1 MHz (2 nos., if available)
5	Additional DC Power Supply	0–5 V (for DC input levels)
6	Digital Storage Oscilloscope	2-channel, 20–50 MHz bandwidth
7	Digital Multimeter	For DC voltage measurements
8	Breadboard and connecting wires	–

Table 7.1: Components and equipment list.

7.4 Theory

Operational amplifiers can perform linear mathematical operations directly on analog voltage signals. Negative feedback forces the differential input voltage to near zero, simplifying circuit analysis through the virtual short/ground approximation.

7.4.1 Inverting Summing Amplifier (Adder)

In an inverting summing configuration, multiple input voltages are connected through individual resistors to the inverting input terminal (pin 2) of the op-amp. The non-inverting terminal (pin 3) is connected to ground through an offset-minimizing resistor R_{comp} .

Because the inverting node is maintained at virtual ground (0 V), Kirchhoff's Current Law (KCL) at the summing junction gives:

$$I_1 + I_2 = I_f \implies \frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} = \frac{0 - V_{out}}{R_f} \quad (7.1)$$

Rearranging for V_{out} :

$$V_{out} = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 \right) \quad (7.2)$$

When all resistors are equal ($R_1 = R_2 = R_f = R$):

$$V_{out} = -(V_1 + V_2) \quad (7.3)$$

The negative sign indicates an overall 180° inversion of the sum.

7.4.2 Difference Amplifier (Subtractor)

A difference amplifier amplifies the potential difference between two input signals while rejecting any common-mode voltage. V_1 is fed to the inverting input via R_1 , and V_2 is fed to the non-inverting input via a voltage divider formed by R_3 and R_4 .

Applying the superposition principle:

1. With $V_2 = 0$ V, the circuit acts as an inverting amplifier: $V_{out1} = -\frac{R_2}{R_1} V_1$.
2. With $V_1 = 0$ V, the voltage at the non-inverting pin is $V_+ = V_2 \left(\frac{R_4}{R_3 + R_4} \right)$. The circuit acts as a non-inverting amplifier:

$$V_{out2} = V_+ \left(1 + \frac{R_2}{R_1} \right) = V_2 \left(\frac{R_4}{R_3 + R_4} \right) \left(\frac{R_1 + R_2}{R_1} \right) \quad (7.4)$$

Combining both contributions yields the net output:

$$V_{out} = V_{out1} + V_{out2} = V_2 \left(\frac{R_4}{R_3 + R_4} \right) \left(\frac{R_1 + R_2}{R_1} \right) - \frac{R_2}{R_1} V_1 \quad (7.5)$$

If $R_1 = R_3$ and $R_2 = R_4$:

$$V_{out} = \frac{R_2}{R_1} (V_2 - V_1) \quad (7.6)$$

For unity differential gain ($R_1 = R_2 = R_3 = R_4 = R$):

$$V_{out} = V_2 - V_1 \quad (7.7)$$

7.5 Circuit Diagram

The schematics for the Inverting Adder and Subtractor are shown in Figures 7.1 and 7.2.

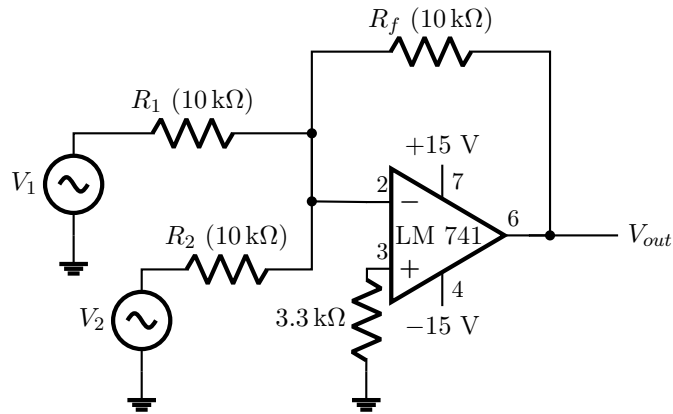


Figure 7.1: Inverting Summing Amplifier (Adder) Circuit Diagram

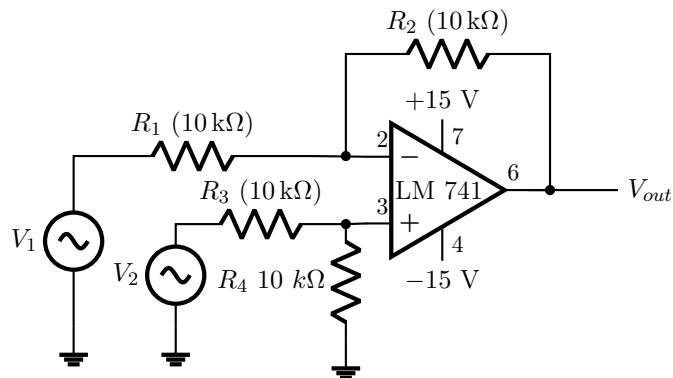


Figure 7.2: Difference Amplifier (Subtractor) Circuit Diagram

7.6 Design

7.6.1 Adder Design

1. Select $R_1 = R_2 = R_f = 10 \text{ k}\Omega$ for unity gain scaling.
2. Calculate R_{comp} to minimize output offset voltage due to input bias currents:

$$R_{comp} = R_1 \parallel R_2 \parallel R_f = \frac{10 \text{ k}\Omega}{3} \approx 3.33 \text{ k}\Omega \quad (\text{Select standard value: } 3.3 \text{ k}\Omega) \quad (7.8)$$

7.6.2 Subtractor Design

1. Select $R_1 = R_2 = R_3 = R_4 = 10 \text{ k}\Omega$ using matched 1% tolerance metal film resistors.
2. Verify that $R_1 \parallel R_2 = R_3 \parallel R_4 = 5 \text{ k}\Omega$, ensuring inherent resistance symmetry at both op-amp input terminals.

7.7 Procedure

7.7.1 Adder Circuit Setup

1. Wire the IC 741 on the breadboard according to Figure 7.1. Connect $\pm 15 \text{ V}$ power supplies.
2. **DC Test:** Apply DC voltages from auxiliary supplies to V_1 and V_2 . Measure V_{out} with a digital multimeter and compare with $-(V_1 + V_2)$.

3. **AC Test:** Apply sinusoidal inputs at 1 kHz: $V_1 = 1.0 V_{p-p}$ and $V_2 = 2 V_{p-p}$ from synchronized generator outputs.¹ Observe V_{out} on the DSO and confirm the sum and 180° phase relation.

7.7.2 Subtractor Circuit Setup

1. Reconfigure the circuit according to Figure 7.2.
2. **DC Test:** Apply DC voltages to V_1 and V_2 . Measure V_{out} using a multimeter and verify $V_{out} = V_2 - V_1$.
3. **AC Test:** Apply two sinusoidal signals at 1 kHz with unequal amplitudes ($V_1 = 1.0 V_{p-p}$, $V_2 = 3.0 V_{p-p}$) in phase². Measure V_{out} on the DSO and verify that $V_{out} = 2 V_{p-p}$.

7.8 Precautions

1. Turn off power before making any circuit modifications.
2. Ensure that the algebraic sum or difference of the input voltages multiplied by the stage gain does not cause V_{out} to exceed the op-amp saturation limits ($\approx \pm 13 V$).
3. Use matched resistors (1% tolerance) in the subtractor circuit to maintain high Common-Mode Rejection Ratio (CMRR).

7.9 Model Graph / Expected Waveforms

Figure 7.3 illustrates the expected AC signal relationships at 1 kHz for both operational modes.

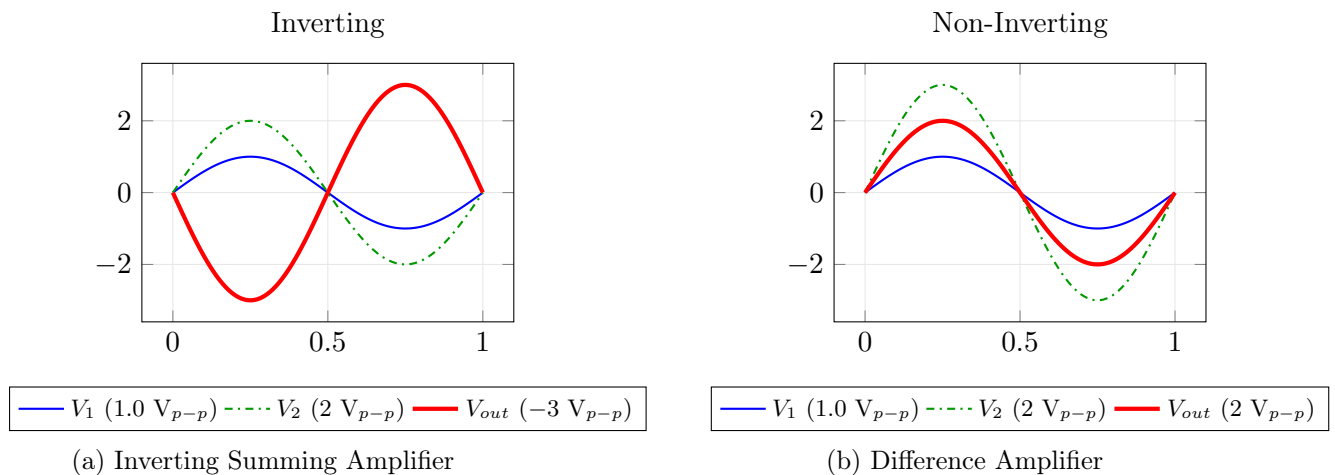


Figure 7.3: Time-Domain Waveforms for Adder and Subtractor

7.10 Observation Tables

7.10.1 Adder Circuit DC Measurements

Record the DC summation observations in Table 7.2.

¹If synchronised signals are not available, V_1 may be obtained from V_2 through a potential divider network.

²If two sources with in-phase signals are not available, you may derive one source from the other using a potential divider network.

v_1 (V)	v_2 (V)	v_3 (V)	v_{out} (V)
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Table 7.2: Summing amplifier: DC observation table (measured vs. Equation (??)).

7.10.2 Subtractor Circuit DC Measurements

Record the DC difference observations in Table 7.3.

v_1 (V)	v_2 (V)	v_{out} (theory)	v_{out} (measured)	% Error
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Table 7.3: Difference amplifier: DC observation table (including the equal-input common-mode test).

7.11 Result

The performance of Op-Amp Adder and Subtractor circuits using IC 741 was successfully evaluated:

- The inverting summing amplifier was verified for DC and AC inputs, confirming that $V_{out} = -(V_1 + V_2)$.
- The difference amplifier was verified, confirming $V_{out} = V_2 - V_1$.

7.12 Questions

1. How can an inverting summing amplifier be modified to compute the mathematical average of three input signals ($V_{out} = -\frac{V_1+V_2+V_3}{3}$)?
2. What happens to the common-mode rejection capability of a subtractor circuit if one of the resistors has a 5% tolerance mismatch?
3. How would you modify the summing amplifier circuit to perform *subtraction* of two signals using only inverting-adder and inverting-amplifier (unity gain) stages, without using the single-op-amp difference amplifier of Figure 7.2?
4. Define Common-Mode Rejection Ratio (CMRR) and explain why a high CMRR is desirable in instrumentation applications.
5. What is a key limitation of the single-op-amp difference amplifier of Figure ?? regarding input impedance, particularly when R_1 is small? How does an instrumentation amplifier (using three op-amps) overcome this limitation?
6. If all three inputs of the summing amplifier in Figure ?? are driven with the same DC voltage V , and $R_1 = R_2 = R_3 = R_f$, what is v_{out} ? Verify this against your observation table.

7. Explain, with an example, one practical application each of a summing amplifier and a difference amplifier in analogue signal processing systems.
8. If $V_1 = 5\text{ V}$ and $V_2 = 10\text{ V}$ are applied to an adder configured with unity gain powered at $\pm 15\text{ V}$, will the output be -15 V ? Explain why or why not.
9. Describe how a non-inverting summing amplifier can be designed using a single op-amp.

Appendix A

Manufacturer Datasheets

NPN general purpose transistors**BC107; BC108; BC109****FEATURES**

- Low current (max. 100 mA)
- Low voltage (max. 45 V).

APPLICATIONS

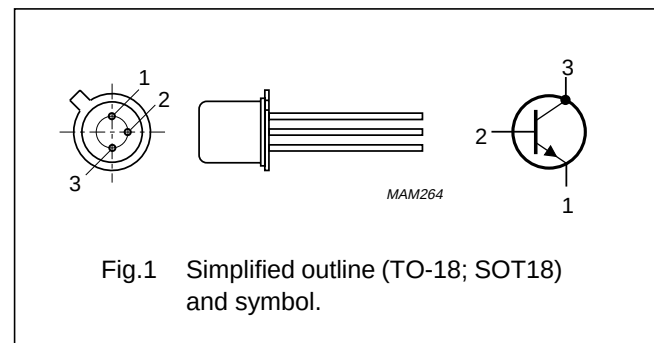
- General purpose switching and amplification.

DESCRIPTION

NPN transistor in a TO-18; SOT18 metal package.
PNP complement: BC177.

PINNING

PIN	DESCRIPTION
1	emitter
2	base
3	collector, connected to the case

**QUICK REFERENCE DATA**

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
V_{CBO}	collector-base voltage BC107 BC108; BC109	open emitter	—	50	V
			—	30	V
V_{CEO}	collector-emitter voltage BC107 BC108; BC109	open base	—	45	V
			—	20	V
I_{CM}	peak collector current		—	200	mA
P_{tot}	total power dissipation	$T_{amb} \leq 25\text{ °C}$	—	300	mW
h_{FE}	DC current gain BC107 BC108 BC109	$I_C = 2\text{ mA}; V_{CE} = 5\text{ V}$	110	450	
			110	800	
			200	800	
f_T	transition frequency	$I_C = 10\text{ mA}; V_{CE} = 5\text{ V}; f = 100\text{ MHz}$	100	—	MHz

NPN general purpose transistors

BC107; BC108; BC109

LIMITING VALUES

In accordance with the Absolute Maximum Rating System (IEC 134).

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
V _{CBO}	collector-base voltage	open emitter			
	BC107		–	50	V
	BC108; BC109		–	30	V
V _{CEO}	collector-emitter voltage	open base			
	BC107		–	45	V
	BC108; BC109		–	20	V
V _{EBO}	emitter-base voltage	open collector			
	BC107		–	6	V
	BC108; BC109		–	5	V
I _C	collector current (DC)		–	100	mA
I _{CM}	peak collector current		–	200	mA
I _{BM}	peak base current		–	200	mA
P _{tot}	total power dissipation	T _{amb} ≤ 25 °C	–	300	mW
T _{stg}	storage temperature		–65	+150	°C
T _j	junction temperature		–	175	°C
T _{amb}	operating ambient temperature		–65	+150	°C

THERMAL CHARACTERISTICS

SYMBOL	PARAMETER	CONDITIONS	VALUE	UNIT
R _{th j-a}	thermal resistance from junction to ambient	note 1	0.5	K/mW
R _{th j-c}	thermal resistance from junction to case		0.2	K/mW

Note

1. Transistor mounted on an FR4 printed-circuit board.

NPN general purpose transistors

BC107; BC108; BC109

CHARACTERISTICS

$T_j = 25\text{ }^{\circ}\text{C}$ unless otherwise specified.

SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
I_{CBO}	collector cut-off current	$I_E = 0; V_{CB} = 20\text{ V}$	–	–	15	nA
		$I_E = 0; V_{CB} = 20\text{ V}; T_j = 150\text{ }^{\circ}\text{C}$	–	–	15	μA
I_{EBO}	emitter cut-off current	$I_C = 0; V_{EB} = 5\text{ V}$	–	–	50	nA
h_{FE}	DC current gain BC107A; BC108A BC107B; BC108B; BC109B BC108C; BC109C	$I_C = 10\text{ }\mu\text{A}; V_{CE} = 5\text{ V}$	–	90	–	
			40	150	–	
			100	270	–	
h_{FE}	DC current gain BC107A; BC108A BC107B; BC108B; BC109B BC108C; BC109C	$I_C = 2\text{ mA}; V_{CE} = 5\text{ V}$	110	180	220	
			200	290	450	
			420	520	800	
V_{CEsat}	collector-emitter saturation voltage	$I_C = 10\text{ mA}; I_B = 0.5\text{ mA}$	–	90	250	mV
		$I_C = 100\text{ mA}; I_B = 5\text{ mA}$	–	200	600	mV
V_{BEsat}	base-emitter saturation voltage	$I_C = 10\text{ mA}; I_B = 0.5\text{ mA}; \text{note 1}$	–	700	–	mV
		$I_C = 100\text{ mA}; I_B = 5\text{ mA}; \text{note 1}$	–	900	–	mV
V_{BE}	base-emitter voltage	$I_C = 2\text{ mA}; V_{CE} = 5\text{ V}; \text{note 2}$	550	620	700	mV
		$I_C = 10\text{ mA}; V_{CE} = 5\text{ V}; \text{note 2}$	–	–	770	mV
C_c	collector capacitance	$I_E = i_e = 0; V_{CB} = 10\text{ V}; f = 1\text{ MHz}$	–	2.5	6	pF
C_e	emitter capacitance	$I_C = i_c = 0; V_{EB} = 0.5\text{ V}; f = 1\text{ MHz}$	–	9	–	pF
f_T	transition frequency	$I_C = 10\text{ mA}; V_{CB} = 5\text{ V}; f = 100\text{ MHz}$	100	–	–	MHz
F	noise figure BC109B; BC109C	$I_C = 200\text{ }\mu\text{A}; V_{CE} = 5\text{ V}; R_S = 2\text{ k}\Omega;$ $f = 30\text{ Hz to }15.7\text{ kHz}$	–	–	4	dB
F	noise figure BC107A; BC108A BC107B; BC108B; BC108C BC109B; BC109C	$I_C = 200\text{ }\mu\text{A}; V_{CE} = 5\text{ V}; R_S = 2\text{ k}\Omega;$ $f = 1\text{ kHz}; B = 200\text{ Hz}$	–	–	10	dB
			–	–	4	dB

Notes

- V_{BEsat} decreases by about 1.7 mV/K with increasing temperature.
- V_{BE} decreases by about 2 mV/K with increasing temperature.



2N7000

N-Channel Enhancement-Mode Vertical DMOS FET

Features

- Free from Secondary Breakdown
- Low Power Drive Requirement
- Ease of Paralleling
- Low C_{ISS} and Fast Switching Speeds
- Excellent Thermal Stability
- Integral Source-Drain Diode
- High Input Impedance and High Gain

Applications

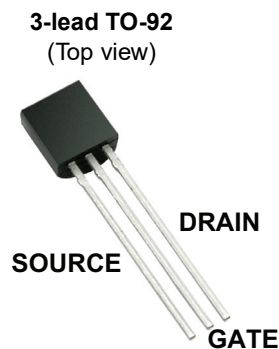
- Motor Controls
- Converters
- Amplifiers
- Switches
- Power Supply Circuits
- Drivers (Relays, Hammers, Solenoids, Lamps, Memory, Displays, Bipolar Transistors, etc.)

General Description

The 2N7000 is an Enhancement-mode (normally-off) transistor that uses a vertical DMOS structure and a well-proven silicon gate manufacturing process. This combination produces a device with the power handling capabilities of bipolar transistors as well as the high input impedance and positive temperature coefficient inherent in MOS devices. Characteristic of all MOS structures, this device is free from thermal runaway and thermally induced secondary breakdown.

Microchip's vertical DMOS FETs are ideally suited to a wide range of switching and amplifying applications where very low threshold voltage, high breakdown voltage, high input impedance, low input capacitance, and fast switching speeds are desired.

Package Type



See [Table 3-1](#) for pin information.

1.0 ELECTRICAL CHARACTERISTICS

Absolute Maximum Ratings†

Drain-to-Source Voltage	BV_{DSS}
Drain-to-Gate Voltage	BV_{DGS}
Gate-to-Source Voltage	$\pm 30V$
Operating Ambient Temperature, T_A	$-55^{\circ}C$ to $+150^{\circ}C$
Storage Temperature, T_S	$-55^{\circ}C$ to $+150^{\circ}C$

† **Notice:** Stresses above those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. This is a stress rating only, and functional operation of the device at those or any other conditions above those indicated in the operational sections of this specification is not intended. Exposure to maximum rating conditions for extended periods may affect device reliability.

DC ELECTRICAL CHARACTERISTICS

Electrical Specifications: $T_A = 25^{\circ}C$ unless otherwise specified. All DC parameters are 100% tested at $25^{\circ}C$ unless otherwise stated. (Pulse test: 300 μs pulse, 2% duty cycle)

Parameter	Sym.	Min.	Typ.	Max.	Unit	Conditions
Drain-to-Source Breakdown Voltage	BV_{DSS}	60	—	—	V	$V_{GS} = 0V$, $I_D = 10 \mu A$
Gate Threshold Voltage	$V_{GS(th)}$	0.8	—	3	V	$V_{GS} = V_{DS}$, $I_D = 1 mA$
Gate Body Leakage Current	I_{GSS}	—	—	10	nA	$V_{GS} = \pm 15V$, $V_{DS} = 0V$
Zero-Gate Voltage Drain Current	I_{DSS}	—	—	1	μA	$V_{GS} = 0V$, $V_{DS} = 48V$
		—	—	1	mA	$V_{GS} = 0V$, $V_{DS} = 48V$, $T_A = 125^{\circ}C$ (Note 1)
On-State Drain Current	$I_{D(ON)}$	75	—	—	mA	$V_{GS} = 4.5V$, $V_{DS} = 10V$
Static Drain-to-Source On-State Resistance	$R_{DS(ON)}$	—	—	5.3	Ω	$V_{GS} = 4.5V$, $I_D = 75 mA$
		—	—	5	Ω	$V_{GS} = 10V$, $I_D = 500 mA$

Note 1: Specification is obtained by characterization and is not 100% tested.

AC ELECTRICAL CHARACTERISTICS

Electrical Specifications: $T_A = 25^{\circ}C$ unless otherwise specified. All AC parameters are not 100% sample tested.

Parameter	Sym.	Min.	Typ.	Max.	Unit	Conditions
Forward Transconductance	G_{FS}	100	—	—	mmho	$V_{DS} = 10V$, $I_D = 200 mA$
Input Capacitance	C_{ISS}	—	—	60	pF	$V_{GS} = 0V$, $V_{DS} = 25V$, $f = 1 MHz$
Common-Source Output Capacitance	C_{OSS}	—	—	25	pF	
Reverse Transfer Capacitance	C_{RSS}	—	—	5	pF	
Turn-On Delay Time	$t_{d(ON)}$	—	—	10	ns	$V_{DD} = 15V$, $I_D = 500 mA$, $R_{GEN} = 25\Omega$
Turn-Off Delay Time	$t_{d(OFF)}$	—	—	10	ns	

DIODE PARAMETER

Diode Forward Voltage Drop	V_{SD}	—	0.85	—	V	$V_{GS} = 0V$, $I_{SD} = 200 mA$ (Note 1)
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Note 1: All DC parameters are 100% tested at $25^{\circ}C$ unless otherwise stated.
(Pulse test: 300 μs pulse, 2% duty cycle)

TEMPERATURE SPECIFICATIONS

Electrical Characteristics: Unless otherwise specified, for all specifications $T_A = T_J = +25^\circ\text{C}$.						
Parameter	Sym.	Min.	Typ.	Max.	Unit	Conditions
TEMPERATURE RANGE						
Operating Ambient Temperature	T_A	-55	—	+150	$^\circ\text{C}$	
Storage Temperature	T_S	-55	—	+150	$^\circ\text{C}$	
PACKAGE THERMAL RESISTANCE						
3-lead TO-92	θ_{JA}	—	132	—	$^\circ\text{C/W}$	

Note 1: Mounted on an FR4 board; 25 mm x 25 mm x 1.57 mm

THERMAL CHARACTERISTICS

Package	I_D (Note 1) (Continuous) (mA)	I_D (Pulsed) (mA)	Power Dissipation at $T_C = 25^\circ\text{C}$ (W)	I_{DR} (Note 1) (mA)	I_{DRM} (mA)
3-lead TO-92	200	500	1	200	500

Note 1: I_D (continuous) is limited by maximum T_J .

2.0 TYPICAL PERFORMANCE CURVES

Note: The graphs and tables provided following this note are a statistical summary based on a limited number of samples and are provided for informational purposes only. The performance characteristics listed herein are not tested or guaranteed. In some graphs or tables, the data presented may be outside the specified operating range (e.g. outside specified power supply range) and therefore outside the warranted range.

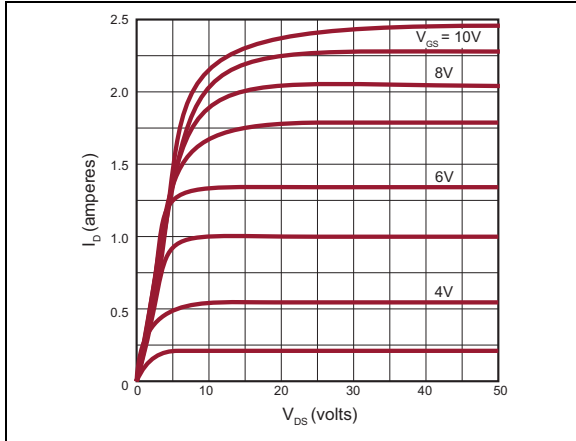


FIGURE 2-1: Output Characteristics.

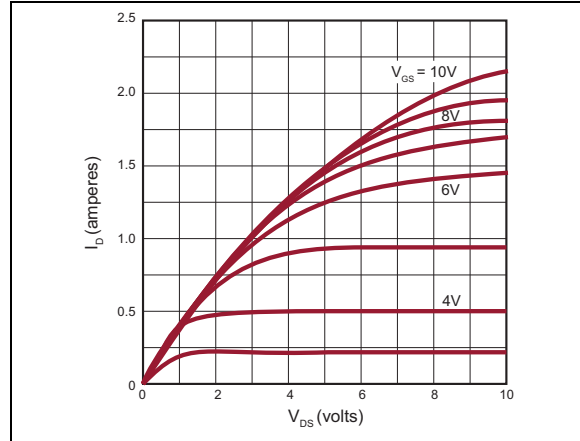


FIGURE 2-4: Saturation Characteristics.

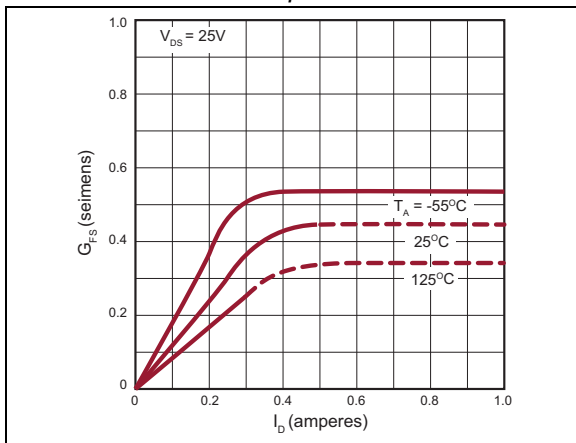


FIGURE 2-2: Transconductance vs. Drain Current.

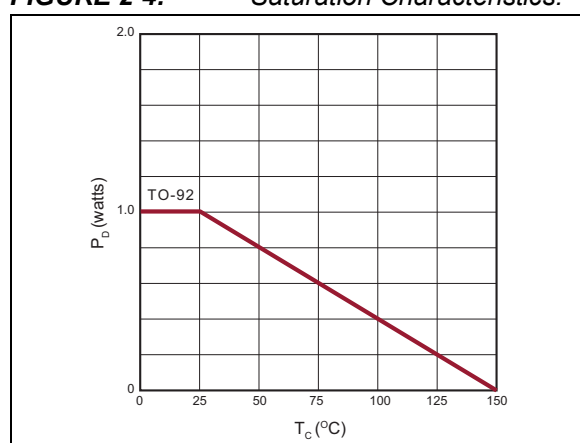


FIGURE 2-5: Power Dissipation vs. Case Temperature.

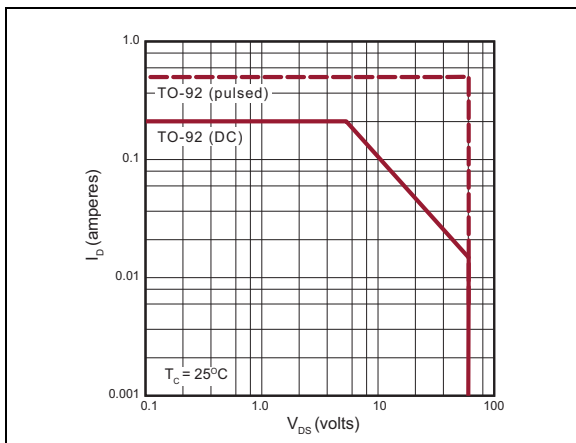


FIGURE 2-3: Maximum Rated Safe Operating Area.

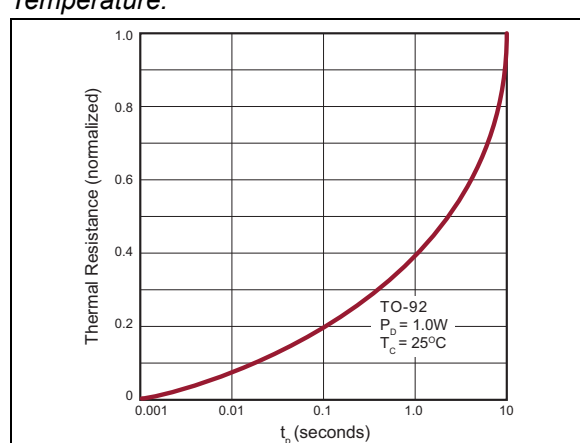


FIGURE 2-6: Thermal Response Characteristics.

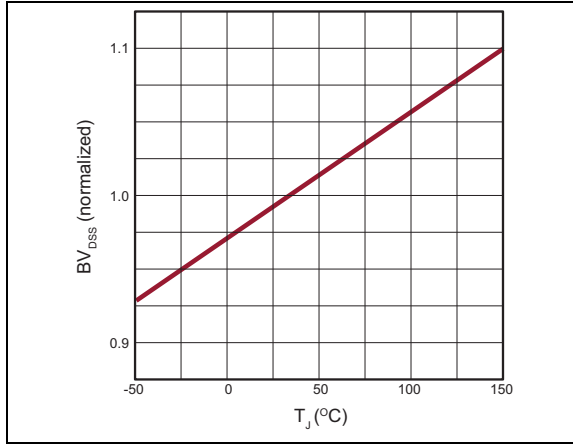


FIGURE 2-7: B_{VDSS} Variation with Temperature.

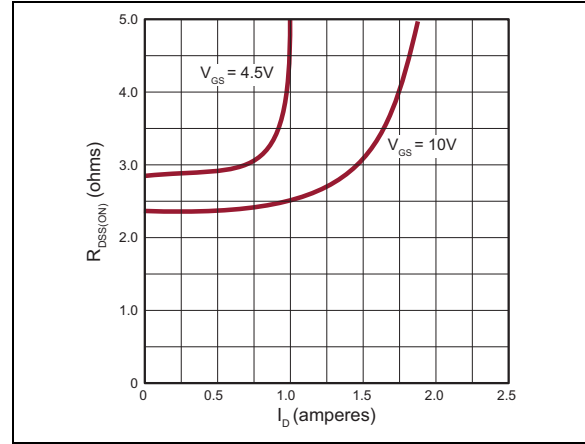


FIGURE 2-10: On-Resistance vs. Drain Current.

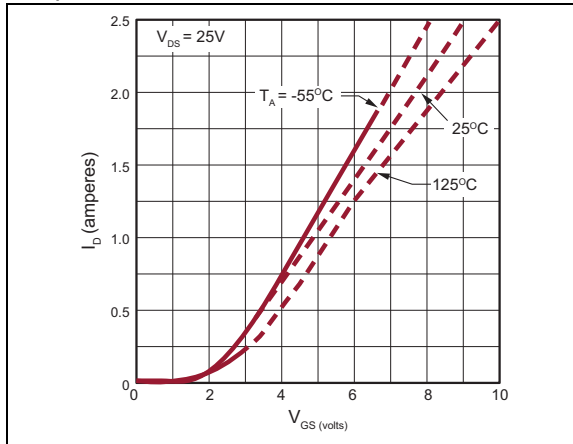


FIGURE 2-8: Transfer Characteristics.

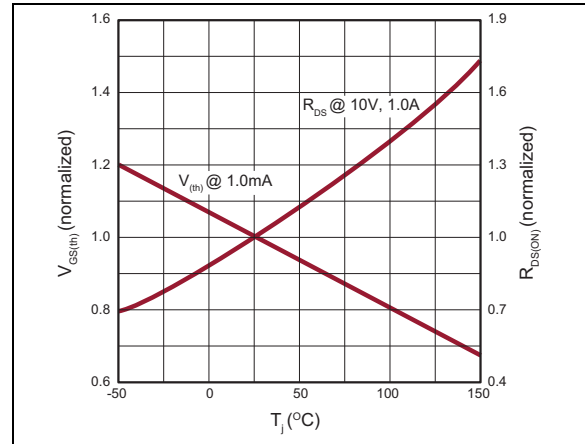


FIGURE 2-11: $V_{GS(th)}$ and R_{DS} Variation with Temperature.

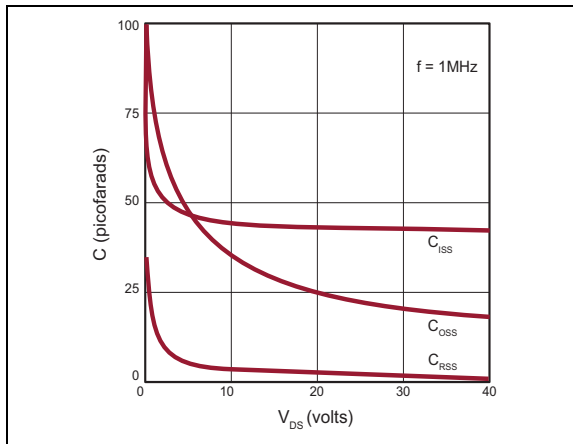


FIGURE 2-9: Capacitance vs. Drain-to-Source Voltage.

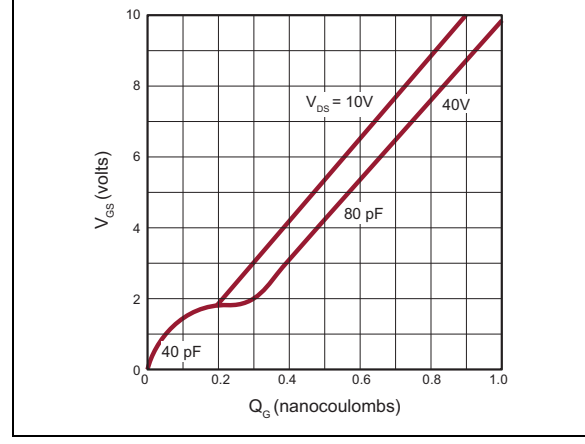
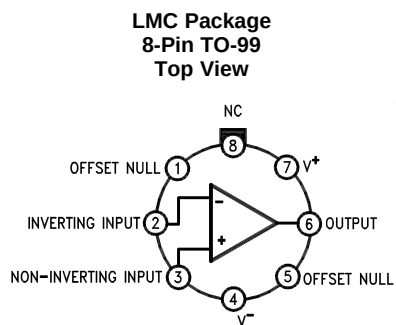
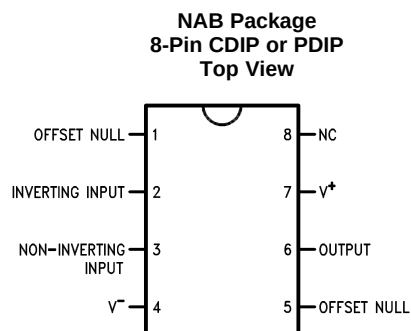


FIGURE 2-12: Gate Drive Dynamic Characteristics.

5 Pin Configuration and Functions



LM741H is available per JM38510/10101



Pin Functions

PIN		I/O	DESCRIPTION
NAME	NO.		
INVERTING INPUT	2	I	Inverting signal input
NC	8	N/A	No Connect, should be left floating
NONINVERTING INPUT	3	I	Noninverting signal input
OFFSET NULL	1, 5	I	Offset null pin used to eliminate the offset voltage and balance the input voltages.
OFFSET NULL			
OUTPUT	6	O	Amplified signal output
V+	7	I	Positive supply voltage
V-	4	I	Negative supply voltage

6 Specifications

6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾⁽²⁾⁽³⁾

		MIN	MAX	UNIT
Supply voltage	LM741, LM741A		±22	V
	LM741C		±18	
Power dissipation ⁽⁴⁾			500	mW
Differential input voltage			±30	V
Input voltage ⁽⁵⁾			±15	V
Output short circuit duration			Continuous	
Operating temperature	LM741, LM741A	–50	125	°C
	LM741C	0	70	
Junction temperature	LM741, LM741A		150	°C
	LM741C		100	
Soldering information	PDIP package (10 seconds)		260	°C
	CDIP or TO-99 package (10 seconds)		300	°C
Storage temperature, T _{stg}		–65	150	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) For military specifications see RETS741X for LM741 and RETS741AX for LM741A.
- (3) If Military/Aerospace specified devices are required, please contact the TI Sales Office/Distributors for availability and specifications.
- (4) For operation at elevated temperatures, these devices must be derated based on thermal resistance, and T_j max. (listed under “Absolute Maximum Ratings”). $T_j = T_A + (\theta_{JA} P_D)$.
- (5) For supply voltages less than ±15 V, the absolute maximum input voltage is equal to the supply voltage.

6.2 ESD Ratings

		VALUE	UNIT
V _(ESD)	Electrostatic discharge Human body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±400	V

- (1) Level listed above is the passing level per ANSI, ESDA, and JEDEC JS-001. JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

		MIN	NOM	MAX	UNIT
Supply voltage (VDD-GND)	LM741, LM741A	±10	±15	±22	V
	LM741C	±10	±15	±18	
Temperature	LM741, LM741A	–55		125	°C
	LM741C	0		70	

6.4 Thermal Information

THERMAL METRIC ⁽¹⁾		LM741			UNIT
		LMC (TO-99)	NAB (CDIP)	P (PDIP)	
		8 PINS	8 PINS	8 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	170	100	100	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	25	—	—	°C/W

- (1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).

6.5 Electrical Characteristics, LM741⁽¹⁾

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
Input offset voltage		$R_S \leq 10\text{ k}\Omega$	$T_A = 25^\circ\text{C}$		1	5	mV
			$T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$			6	mV
Input offset voltage adjustment range		$T_A = 25^\circ\text{C}$, $V_S = \pm 20\text{ V}$			± 15		mV
Input offset current		$T_A = 25^\circ\text{C}$			20	200	nA
		$T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$			85	500	
Input bias current		$T_A = 25^\circ\text{C}$			80	500	nA
		$T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$				1.5	μA
Input resistance		$T_A = 25^\circ\text{C}$, $V_S = \pm 20\text{ V}$			0.3	2	M Ω
Input voltage range		$T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$			± 12	± 13	V
Large signal voltage gain		$V_S = \pm 15\text{ V}$, $V_O = \pm 10\text{ V}$, $R_L \geq 2\text{ k}\Omega$	$T_A = 25^\circ\text{C}$		50	200	V/mV
			$T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$		25		
Output voltage swing		$V_S = \pm 15\text{ V}$	$R_L \geq 10\text{ k}\Omega$		± 12	± 14	V
			$R_L \geq 2\text{ k}\Omega$		± 10	± 13	
Output short circuit current		$T_A = 25^\circ\text{C}$			25		mA
Common-mode rejection ratio		$R_S \leq 10\text{ }\Omega$, $V_{\text{CM}} = \pm 12\text{ V}$, $T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$			80	95	dB
Supply voltage rejection ratio		$V_S = \pm 20\text{ V}$ to $V_S = \pm 5\text{ V}$, $R_S \leq 10\text{ }\Omega$, $T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$			86	96	dB
Transient response	Rise time	$T_A = 25^\circ\text{C}$, unity gain			0.3		μs
	Overshoot				5%		
Slew rate		$T_A = 25^\circ\text{C}$, unity gain			0.5		V/ μs
Supply current		$T_A = 25^\circ\text{C}$			1.7	2.8	mA
Power consumption		$V_S = \pm 15\text{ V}$	$T_A = 25^\circ\text{C}$		50	85	mW
			$T_A = T_{\text{AMIN}}$		60	100	
			$T_A = T_{\text{AMAX}}$		45	75	

(1) Unless otherwise specified, these specifications apply for $V_S = \pm 15 \text{ V}$, $-55^\circ\text{C} \leq T_A \leq +125^\circ\text{C}$ (LM741/LM741A). For the LM741C/LM741E, these specifications are limited to $0^\circ\text{C} \leq T_A \leq +70^\circ\text{C}$.

6.6 Electrical Characteristics, LM741A⁽¹⁾

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
Input offset voltage	$R_S \leq 50 \Omega$	$T_A = 25^\circ\text{C}$		0.8	3	mV
		$T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$			4	mV
Average input offset voltage drift					15	$\mu\text{V}/^\circ\text{C}$
Input offset voltage adjustment range	$T_A = 25^\circ\text{C}$, $V_S = \pm 20 \text{ V}$		± 10			mV
Input offset current	$T_A = 25^\circ\text{C}$			3	30	nA
	$T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$				70	
Average input offset current drift					0.5	nA/ $^\circ\text{C}$
Input bias current	$T_A = 25^\circ\text{C}$			30	80	nA
	$T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$				0.21	μA
Input resistance	$T_A = 25^\circ\text{C}$, $V_S = \pm 20 \text{ V}$		1	6		M Ω
	$T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$, $V_S = \pm 20 \text{ V}$		0.5			
Large signal voltage gain	$V_S = \pm 20 \text{ V}$, $V_O = \pm 15 \text{ V}$, $R_L \geq 2 \text{ k}\Omega$	$T_A = 25^\circ\text{C}$	50			V/mV
		$T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$	32			
	$V_S = \pm 5 \text{ V}$, $V_O = \pm 2 \text{ V}$, $R_L \geq 2 \text{ k}\Omega$, $T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$		10			

(1) Unless otherwise specified, these specifications apply for $V_S = \pm 15 \text{ V}$, $-55^\circ\text{C} \leq T_A \leq +125^\circ\text{C}$ (LM741/LM741A). For the LM741C/LM741E, these specifications are limited to $0^\circ\text{C} \leq T_A \leq +70^\circ\text{C}$.